

SingleRAN

OM Security Feature Parameter Description

Issue	Draft A
Date	2025-12-31



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Contents

1 Change History.....	1
1.1 SRAN22.1 Draft A (2025-12-31).....	1
2 About This Document.....	2
2.1 General Statements.....	2
2.2 Applicable RAT.....	2
2.3 Features in This Document.....	3
3 Overview.....	4
4 Security Management.....	6
4.1 Principles.....	6
4.1.1 User Management.....	6
4.1.1.1 Overview.....	6
4.1.1.2 Login Authentication.....	8
4.1.1.3 User Rights Control.....	9
4.1.1.4 Login Password Policy.....	12
4.1.1.5 Simultaneous Online User Number Management.....	14
4.1.1.6 Southbound Interface Access Management.....	16
4.1.1.7 FTP User Management.....	17
4.1.2 OMCH Security.....	18
4.1.2.1 SSL-Encrypted Transmission.....	18
4.1.2.2 SSH-Encrypted Transmission.....	19
4.1.2.3 Management-Plane IP Address Isolation.....	21
4.1.2.4 Management Plane Isolation by VRFs.....	22
4.1.2.5 Authentication between the EMS and NEs.....	22
4.1.2.6 OMU Anti-Attack for the BSC6900/BSC6910.....	22
4.1.3 Security Configuration Check.....	23
4.1.3.1 Overview.....	23
4.1.3.2 Check Baseline Management.....	25
4.1.3.3 Check Template Management.....	25
4.1.3.4 Check Task Management.....	26
4.1.3.5 Configuration Assessment.....	26
4.1.4 Application Security.....	27
4.1.4.1 Web Security.....	27

4.1.4.1.1 Overview.....	27
4.1.4.1.2 HTTPS-based Data Transmission.....	27
4.1.4.1.3 Session Management.....	27
4.1.4.1.4 Anti-attack.....	28
4.1.4.2 Time Security.....	29
4.1.4.2.1 SNTP Security for the BSC6900/BSC6910 (GSM/UMTS).....	29
4.1.4.2.2 NTP Security Authentication for the Base Station/BSC6960.....	29
4.1.5 Security Monitoring.....	30
4.1.5.1 Security Monitoring.....	30
4.1.5.2 Security Alarms, Events, and Logs.....	31
4.1.5.2.1 Overview.....	31
4.1.5.2.2 Security Alarms and Events.....	31
4.1.5.2.3 Security Logs and Security Audit.....	33
4.1.5.2.4 NE Resource Monitoring.....	38
4.1.6 Data Security.....	38
4.1.6.1 Personal Data Security.....	38
4.1.6.1.1 User Identity Security Processing.....	38
4.1.6.1.2 Personal Data Protection.....	39
4.1.6.2 Security Management of Configuration Files.....	39
4.1.6.2.1 Overview.....	39
4.1.6.2.2 Application Scenarios.....	39
4.1.6.2.3 Configuration File Encryption.....	40
4.1.6.3 Digital Signature-based Software Integrity Protection.....	41
4.1.6.3.1 Definition.....	41
4.1.6.3.2 Application Scenarios.....	41
4.1.6.3.3 Software Digital Signature.....	41
4.1.6.3.4 Possible Issues.....	45
4.1.6.4 Configuration Security Management.....	47
4.2 Network Analysis.....	47
4.2.1 Benefits.....	47
4.2.2 Impacts.....	48
4.3 Requirements.....	48
4.3.1 Licenses.....	48
4.3.2 Functions.....	48
4.3.2.1 Prerequisite Functions.....	48
4.3.2.2 Mutually Exclusive Functions.....	48
4.3.2.3 Function Impacts.....	48
4.3.3 Hardware.....	48
4.3.4 Others.....	49
4.4 Operation and Maintenance.....	49
4.4.1 User Management.....	49
4.4.1.1 Data Configuration.....	49

4.4.1.1.1 Using MML Commands.....	49
4.4.1.1.2 Using the MAE-Deployment.....	51
4.4.1.2 Activation Verification.....	51
4.4.1.3 Network Monitoring.....	51
4.4.2 OMCH Security.....	51
4.4.2.1 OMU Anti-Attack for Base Station Controllers.....	51
4.4.2.1.1 When to Use.....	51
4.4.2.1.2 Data Configuration.....	52
4.4.2.1.3 Activation Verification.....	53
4.4.2.1.4 Network Monitoring.....	53
4.4.2.2 SSH-Encrypted Transmission.....	53
4.4.2.2.1 When to Use.....	53
4.4.2.2.2 Data Configuration.....	53
4.4.2.2.3 Activation Verification.....	54
4.4.2.2.4 Network Monitoring.....	54
4.4.3 Security Configuration Check.....	54
4.4.3.1 When to Use.....	54
4.4.3.2 Data Configuration.....	54
4.4.3.3 Activation Verification.....	55
4.4.3.4 Network Monitoring.....	55
4.4.4 Application Security.....	55
4.4.4.1 Web Security.....	55
4.4.4.1.1 When to Use.....	55
4.4.4.1.2 Data Configuration.....	55
4.4.4.1.3 Activation Verification.....	57
4.4.4.1.4 Network Monitoring.....	57
4.4.4.2 Time Security.....	57
4.4.4.2.1 SNTP Security for the BSC6900/BSC6910 (GSM/UMTS).....	57
4.4.4.2.2 Deployment of NTP Security Authentication for the Base Station/BSC6960.....	58
4.4.5 Security Monitoring.....	59
4.4.5.1 Security Monitoring.....	59
4.4.5.2 Security Alarms, Events, and Logs.....	60
4.4.6 Data Security.....	60
4.4.6.1 User Data Pseudonymization.....	60
4.4.6.1.1 When to Use.....	60
4.4.6.1.2 Data Configuration.....	60
4.4.6.1.3 Activation Verification.....	62
4.4.6.1.4 Network Monitoring.....	62
4.4.6.2 Security Management of Configuration Files.....	62
4.4.6.2.1 When to Use.....	62
4.4.6.2.2 Data Configuration.....	62
4.4.6.2.3 Activation Verification.....	63

4.4.6.2.4 Network Monitoring.....	63
4.4.6.3 Digital Signature-based Software Integrity Protection.....	63
4.4.6.4 Configuration Security Management.....	64
4.4.6.4.1 When to Use.....	64
4.4.6.4.2 Data Configuration.....	64
4.4.6.4.3 Activation Verification.....	67
4.4.6.4.4 Network Monitoring.....	68
5 Parameters.....	69
6 Counters.....	71
7 Glossary.....	72
8 Reference Documents.....	73

1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes
Changes in functions and their corresponding parameters
- Editorial changes
Improvements or revisions to the documentation

1.1 SRAN22.1 Draft A (2025-12-31)

This issue introduces the following changes to SRAN21.1 04 (2025-08-30).

Technical Changes

Change Description	Parameter Change	Base Station Model
Added support for security management by UMPThb series main control boards. For details, see Boards .	None	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Editorial Changes

Revised descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in this document apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Applicable RAT

This document applies to GSM, UMTS, LTE FDD, LTE TDD, NB-IoT, and NR. This feature works the same way with each of these RATs.

For definitions of base stations described in this document, see section "Base Station Products" in *SRAN Networking and Evolution Overview*.

2.3 Features in This Document

This document describes the following features.

RA T	Feature ID	Feature Name	Chapter/Section
UM TS	MRFD-210305	Security Management	4 Security Management
LTE FD D	LBFD-004010	Security Management	
LTE TD D	TDLBFD-004010	Security Management	
NB- IoT	MLBFD-12000410	Security Management	

3 Overview

The following table lists the O&M security measures supported by Huawei network elements (NEs).

Table 3-1 Supported security measures

Security Measures		MBSC	BSC6960	eGBTS	NodeB/ eNodeB/ gNodeB/ Multimode Base Station
User management		√	√	√	√
OMCH security		√	√	√	√
Security configuration check		√	√	√	√
Applic ation securi ty	Web security	√	√	√	√
	Time security	√	√	√	√
Securi ty monit oring	Security monitoring	√	√	x	√
	Security alarms, events, and logs	√	√	√	√
Data securi ty	Personal data security	√	√	√	√
	Security management of configuration files	√	√	√	√

Security Measures		MBSC	BSC6960	eGBTS	NodeB/ eNodeB/ gNodeB/ Multimode Base Station
	Digital signature-based software integrity protection	√	√	√	√
	Configuration security management	√	x	√	√

 NOTE

In the preceding tables, √ indicates that the NE supports this security measure. x indicates that the NE does not support this security measure. - indicates that the NE does not involve this security measure.

In this document, eGBTS, NodeB, eNodeB, gNodeB, and MBTS are all referred to as the base station. For details about GBTS OM security, see *GBTS Equipment and OM Security* in *GBSS Feature Documentation*.

4 Security Management

4.1 Principles

4.1.1 User Management

4.1.1.1 Overview

Table 4-1 describes user management functions, which implement authentication and access control of users.

Authentication and access control are performed on users who are attempting to log in to an NE for OM. Authentication identifies users, and access control defines and restricts the operations that users can perform and the resources they can access.

Table 4-1 User management functions

Function	Description
User account management	• Adding^a, modifying, deleting, and disabling accounts • Querying account information, including whether a default password is still being used for an account
User password management	• Restricting the minimum password length and enforcing password complexity • Defining the maximum password validity period • Prohibiting the reuse of recent passwords

Function	Description
Login management ^b	• Authenticating a user identity based on the account and password. • Specifying a valid login period • Requesting verification codes and preventing brute-force cracking by locking user accounts with successive login failures • Displaying information such as the time and IP address of last login and indicating whether a default password^c is being used after a successful login • Locking the GUI if no operation is performed within a specified period of time
User operation authentication	• Authenticating operation objects • Authenticating operation NEs • Restricting operation GUIs • Restricting the commands that users can execute • Restricting directories that users can access (over FTP or through the File Manager tab page of the WebLMT) • Specifying message tracing permission
Centralized user monitoring	• Monitoring online user status • Monitoring user operations • Forcing users out
Centralized user management	• Authenticating users in a centralized manner using the element management system (EMS) • Delivering or revoking rights of domain users^d • Degrading local user^e management • Synchronizing local user management policies

Function	Description
	a: SRAN16.1 allows for a maximum of 50 local user accounts. An earlier version supports a maximum of 10 local user accounts. When a rollback is performed, the accounts that were created in the earlier version and have not been deleted will be retained. However, if more than 10 accounts exist in the new version, the accounts that were added in the new version may be lost. To query the retained accounts, run the LST OP command. b: To log in to the OMU of a base station controller for OM, you can log in as user lgnusr and then switch to user root for performing related operations as required. c: The default password of the system includes the password pre-configured before delivery, password of a new user, and password of a local user changed by administrators. d: Domain users perform routine OM and are centrally managed by the MAE. They are created, modified, authenticated, and authorized by the MAE. Domain users having the MOD OP command permission can run this command to change the password of user admin. e: Local users participate in site deployment and perform local OM in the event of transmission faults.

4.1.1.2 Login Authentication

User login authentication on an NE (base station controller, or base station) involves the following types of users:

- Local users
- Domain users

Local users manage NE configurations using the WebLMT. Domain users manage NE configurations using the MAE. A domain user is also allowed to log in to the WebLMT to access an NE. In this case, the NE forwards a user account and password authentication request to the MAE for user identity authentication.

Controlling Login Time

The following login time control policies are used to more effectively manage login users to ensure access security:

- Setting account validity periods. After the validity period of an account elapses, the account is automatically disabled. In this case, users are allowed to request the security administrator to extend the account validity period and resume the account validity.
- Setting permissible access time ranges for user accounts. The ranges include validity date ranges, time ranges, and specific days of a week. Login is not allowed beyond the permissible access time ranges. The security administrator can adjust the permissible access time ranges.

Displaying Login Status

Users are prompted with login status to indicate security risks. If there is a login failure, no excessively elaborated login failure information is displayed.

Locking Non-default Accounts

Administrators are allowed to run the **SET OPLOCK** command to lock or unlock local non-default user accounts in batches. After the policy of locking insecure accounts is used, local non-default user accounts cannot be used for logging in to network devices. When network devices are disconnected from the MAE and administrator accounts are also locked, the policy becomes invalid. In this case, non-default user accounts are allowed to access the network devices through the LMT in emergencies.

Monitoring Users

The MAE allows for query of online local and domain users and monitoring of their status (login or logout).

The MAE allows for monitoring of all operations of specified online users. When detecting that users are forcibly logged out, the MAE forcibly disrupts the connections of these users. The MAE determines the rights for user monitoring. The base station controller or base station determines the users to be monitored according to the commands received from the MAE and reports the results to the MAE.

Local Login Events of Users

A local login indicates that a local or domain user logs in to the base station through the WebLMT. However, the MAE and north-bound system are unaware of user login in real time, resulting in security risks.

To address this issue, the base station generates an event upon user login to notify the MAE and north-bound system in real time. Users are allowed to subscribe to the event through the north-bound system and check local login information immediately after receiving the alarm.

Only base stations support generation of local login events.

Automatic Disabling of Accounts Not Used for a Long Time

Base stations can automatically disable local non-administrator accounts that are not used for a long time. After an account is disabled, the administrator has permissions to enable the account. This function is enabled by default and is automatically enabled after an upgrade. By default, when an account is not used for 90 days, the account disabling is triggered. The administrator is allowed to run the **SET OPSW** command to enable or disable this function and modify the duration for identifying an idle account.

4.1.1.3 User Rights Control

The base station/base station controller defines five user levels: Administrator(s), Operator(s), User(s), Guest(s), and Custom(s). Rights of these users to use command groups are defined as follows:

- The rights of Administrator(s), Operator(s), User(s), and Guest(s) to use command groups are fixed.
- The rights of Custom(s) to use command groups are defined depending on actual requirements.

A command group is a group of commands that have the same attributes, for example, the G_8 command group. The **LST CCG** command can be used to query the specific commands in a command group.

To query the BSC6900/BSC6910 accounts that are authorized to execute a command on GSM and UMTS networks, perform the following steps:

1. Run the **LST CMDVEST** command to query the default and user-defined command groups that contain a target command.
2. Run the **LST OP** command to query the accounts that are authorized to execute these command groups.

For a base station/BSC6960, run the **LST CMDS** command to query the MML commands that can be executed by the current user.

Table 4-2 and **Table 4-3** list the mapping between user levels and command groups.

Table 4-2 Mapping between user levels and command groups on the BSC6900/BSC6910

User Level	Command Group
Administrator(s)	G_0, G_1, G_2, G_3, G_4, G_5, G_6, G_7, G_8, G_9, G_10, G_11, G_12, G_13, G_14, G_OTHER
Operator(s)	G_0, G_2, G_3, G_4, G_5, G_6, G_7, G_8, G_9, G_10, G_11, G_12, G_13, G_14, G_OTHER
User(s)	G_0, G_2, G_4, G_6, G_7, G_8, G_9, G_10, G_11, G_12, G_13, G_14, G_OTHER
Guest(s)	G_0, G_2, G_4, G_6, G_8, G_13, G_OTHER
Custom(s)	To be added by the user

Table 4-3 Mapping between user levels and command groups on the base station/BSC6960

User Level	Command Group
Administrator(s)	G_0, G_1, G_2, G_3, G_4, G_5, G_6, G_7, G_8, G_9, G_10, G_11, G_12, G_13, G_14, G_15, G_16, G_17, G_18, G_19, G_20, G_21
Operator(s)	G_0, G_2, G_3, G_4, G_5, G_6, G_7, G_8, G_10, G_11, G_12, G_13, G_16, G_17, G_18, G_19, G_20, G_21

User Level	Command Group
User(s)	G_0, G_2, G_3, G_4, G_5, G_6, G_7, G_8, G_10, G_11, G_12, G_13, G_16, G_17, G_18, G_19, G_20, G_21
Guest(s)	G_0, G_2, G_4, G_6, G_8, G_10, G_12, G_16, G_18, G_20
Custom(s)	To be added by the user

Users can perform operations only after a successful login. All user operations are monitored and operation permission is controlled. All operations must be classified according to permission levels.

User operation permission is controlled by running MML commands or performing WebLMT menu operations. Each MML command or menu item can be associated with a command group. The base station controller supports authorizing users to use command groups.

Before users perform operations on NEs and objects or run commands, the system checks their operation permission levels to determine whether the operations are allowed. When users perform operations beyond their permission, the system prompts them with a message, indicating that the operations cannot be performed.

User permission information is stored on servers. After users successfully log in to the clients, the servers send user permission lists to the clients. The user permission lists are always stored on clients before users log out.

The system does not allow users to run any commands beyond permissible time ranges.

If required, administrators can grant permission to a specific user. If users attempt to access a base station controller beyond the permissible time range, the base station controller refuses to perform user authentication. If users use expired passwords for login, the system forces users to change their passwords.

Accessing the Web Server Directory Using the BSC6900/BSC6910 File Manager (GSM/UMTS)

Each user that uses the WebLMT can download or upload files on the **File Manager** tab page. Different levels of users have different rights to obtain information:

User Level	Download Files	Upload Files	Delete Files
Administrator(s)	√	√	√
Operator(s)	√	√	x
User(s)	√	x	x
Guest(s)	√	x	x

Custom(s)	User-defined	User-defined	User-defined
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On the BSC6900/BSC6910, file transfer and access control rules are listed in the output of the **LST FTPDIRAUTH** command.

In addition, to prevent the leakage of the sensitive information on the OMU and the upload of malicious files to the OMU, administrators need to configure file transfer and access control rules based on the user levels by performing the following steps:

- Run the **ADD FTPACCCTRL** command to add file transfer and access control rules.
- Run the **RMV FTPACCCTRL** command to remove unnecessary file transfer and access control rules.
- Run the **LST FTPACCCTRL** command to query file transfer and access control rules.

File transfer and access control rules take effect for files to be uploaded to or downloaded from the FTP server of the BSC6900/BSC6910 or the **File Manager** function of the WebLMT.

Performing Operations on the WebLMT GUI

Local Custom(s) users can be authorized based on function items.

4.1.1.4 Login Password Policy

The **PWDPOLICY** MO can be configured to specify login password policies. For complete login password policies of an NE, see the **SET PWDPOLICY** command help of the corresponding NE. The key login password policies are as follows:

- Password Minimal Length
- Password Complicacy
- Password Max Miss Times
- New Password Repeat Limit
- Password Validity
- Must Modify Password When First Login Switch
- Weak Password Dictionary Check

The default minimum length of a password for a base station/BSC6960 is 12 characters. The actual value range of the minimum password length specified for a base station is 8 to 64 characters.

The default password for a base station/BSC6960 is **Y5\$7Kc@u#\$tr&LhF** (case-sensitive).

NOTICE

Weak password dictionary check is enabled by default and is not under parameter control.

You are advised to change the password according to the latest default password security policy after the base station is upgraded to SRAN16.1.

If the length of a password configured for a base station is greater than 32 characters and the base station is rolled back to a version earlier than SRAN16.1, the login to the rollback target version using the current user account will fail.

Password Usage Rules

To ensure that passwords are not disclosed, tampered with, or stolen, the system adheres to the following password usage rules:

- Passwords entered are displayed as asterisks (*).
- New user passwords must be verified when they are created, and copying passwords is prohibited.
- Old passwords must be verified when the passwords are changed.
- During password change, administrators can only reset the passwords of target users but cannot view their plaintext passwords.
- User accounts are locked when the number of consecutive password attempts has reached a specified threshold.

Password Storage and Transmission Rules

The system adheres to the following password storage and transmission rules:

- Passwords are encrypted using irreversible algorithms when stored locally.
- Administrators cannot retrieve passwords in the form of plaintext or query other users' passwords.

Password Validity Period Management

The system manages password validity periods using the following methods:

- Users are forced to change their passwords upon expiration.
- Users are forced to change the automatically allocated default passwords upon first login.
- Users are prompted to change their passwords before expiration (unless otherwise password expiration alerts are disabled by administrators on the MAE). If passwords are not changed after expiration, users cannot log in to the system using the passwords, but the passwords can be changed or reset on the MAE.

Weak Password Dictionary Management

Weak password dictionaries include the system default weak password dictionary and user-defined weak password dictionary. The system default weak password dictionary is released with the version. The user-defined weak password dictionary

is a supplement to the system default weak password dictionary. A user-defined weak password dictionary can be imported to the NE and activated using commands.

After the weak password dictionary management function is enabled, users cannot use the default password of a base station as a new password during password creation or modification.

The methods of checking weak password dictionaries supported by the system include the following:

On the base station/BSC6960 side

- Run the **DLD WKPWDDICT** command to download a user-defined weak password dictionary file to the base station. This file is used for weak password verification.
- Run the **ULD WKPWDDICT** command to upload the user-defined weak password dictionary file and default weak password dictionary file of the system to an FTP server. The user-defined weak password dictionary file is named **usrdictfile.txt**. The system default weak password dictionary file is named **sysdictfile.txt**.

On the BSC6900/BSC6910 side (GSM/UMTS)

- Run the **DLD USRWKPWDDICT** command to download a user-defined weak password dictionary file to the OMU. A base station controller supports only one user-defined weak password dictionary file. If multiple files have been downloaded, the last file downloaded to the OMU takes effect.
- Run the **ULD USRWKPWDDICT** command to upload a user-defined weak password dictionary file to an FTP server.
- Run the **LST USRWKPWDSTAT** command to query the status of the configured user-defined weak password dictionary file.
- Run the **RMV USRWKPWDDICT** command to remove a user-defined weak password dictionary file from the OMU. After the command execution, the user-defined weak password dictionary file no longer takes effect. The user-defined weak password dictionary file is named **usrweakdict.txt**. The weak password dictionary file contained in the OMU software package is **WKWORDS.dict**.

4.1.1.5 Simultaneous Online User Number Management

Concepts

- Number of online instances
A login instance is added each time a local user or domain user successfully logs in to an NE through the WebLMT. This login instance is available until the user logs out.
A single user can be allocated multiple login instances through repeated login. The total number of login instances of all users is referred to as the number of online instances on an NE.
If five users use the same administrator account to successfully log in to an NE, each successful login is allocated a login instance, that is, the number of online instances is five.

- **Maximum number of online instances**
Each login instance of an NE occupies system resources. The maximum number of online instances is predefined, but not configurable. For example, the base station controller allows a maximum of 32 online instances and a co-MPT base station allows a maximum of 6 online instances.
Specifically, when the number of online instances on the base station controller has reached 32, other users cannot log in to the base station controller until any users log out.

Simultaneous Online User Number Management

Simultaneous online user number management is used to control the maximum login instances of a user on an NE, thereby ensuring that multiple users can concurrently log in to an NE. Without this function, one or more users may repeatedly log in to an NE and do not log out, preventing other local users from login when the number of online instances reaches the maximum allowed by the NE and affecting the O&M of the NE.

This function can be configured using the **SET USRMAXONLINE** command.

Configuration Type in this command can be set to any of the following values:

- **LOCAL_USER_GENERAL(General Configuration of Local Users):** The maximum number of online instances is set to the same value for all local users.
For example, when **Max Users Online** is set to **3**, new login request of any local user with three online instances is denied.
- **SPECIFIED_LOCAL_USER(Configuration of a Specified Local User):** The maximum number of online instances is specific to a local user.
For a local user, this configuration takes precedence over the preceding general configuration.
- **DOMAIN_USER_GENERAL(General Configuration of Domain Users):** The maximum number of online instances is set to the same value for all domain users.
For example, when **Max Users Online** is set to **3**, new login request of any domain user with three online instances is denied.
- **RESTORE_ALL_LOCAL_USER(Restore to General for All Local Users):** The maximum number of online instances for all local users is restored to the value specified when **Configuration Type** is set to **LOCAL_USER_GENERAL(General Configuration of Local Users)**.
- **RESTORE_SPECIFIED_LOCAL_USER(Restore to General for One Local User):** The maximum number of online instances for a specified local user is restored to the value specified when **Configuration Type** is set to **LOCAL_USER_GENERAL(General Configuration of Local Users)**.

The **LST USRMAXONLINE** command can be used to query the configurations, including general configuration for local users, general configuration for domain users, and the maximum number of online instances for a specified local user

It is good practice to set the maximum number of online instances as follows:

- Set the maximum number to **1** for users of the administrator level, including the admin user, thereby enhancing system security.

- Set the maximum number based on the number of admitted terminals and tools for accounts used by the terminals or tools.

The restrictions on the total number of online instances apply to both users and login systems. If the total number of online instances of all online users reaches the upper limit allowed by the login system, other users cannot log in until any online user logs out.

4.1.1.6 Southbound Interface Access Management

The MAE and NetEco use pre-shared keys for identity authentication when connecting to an NE over the southbound interface. To distinguish between OSS types, the MAE and NetEco use the **emscomm** and **emscommneteco** accounts as their respective identities. **ei*b+@b#6Nh(ts1j** is the default password for both **emscomm** and **emscommneteco**. Specific OSSs can create accounts for temporary login to the base station during automatic OMCH establishment. The account name is specified by the user and the default password is **)@YEufV6tb%R5((=**.

NOTE

The trace server (TS) and fault assistant resolved solution (FARS) are subsystems of the MAE and use the MAE's identity credentials to access NEs. Generally, the identity credentials are not distinguished between the MAE and subsystems (TS or FARS) in NE logs. However, in some base station logs, TS and FARS are identified with the **emscommmts** and **emscommfars** accounts, respectively.

If password authentication fails for three consecutive times for the OSS account **emscomm** or **emscommneteco** that is used to log in to the base station, the base station starts the delayed login function. That is, any login attempt is prohibited within 1 minute after each login failure. The base station stops this function when the login succeeds or the password is changed.

The **emscommneteco** account is disabled by default.

The password for the account must be consistent between an NE and the OSS. Otherwise, the NE cannot connect to the OSS.

MAE

On the MAE-Access, different passwords of the **emscomm** account can be configured for different NEs. Users can choose **Security > Set emscomm User Password** on the MAE-Access to change the password of the **emscomm** account on the NE and the MAE simultaneously.

In addition, users must plan the global **emscomm** password in advance. In new MAE deployment and MAE upgrade scenarios, the password is forcibly set to replace the default password preset before delivery for wireless NEs (including base stations and base station controllers). When a new NE is connected to the MAE for the first time or an existing NE is upgraded, the MAE automatically changes the **emscomm** password for the NE to the global password planned by users.

When an NE is disconnected from the OSS (for example, during NE board replacement), and the cause of the disconnection alarm is displayed as login failure on the OSS, perform the following steps:

- On the NE side

- a. Use a local administrator account to log in to the LMT of the NE by using the MAE-Access proxy.
 - b. Run the **MOD OP** command to change the password of the **emscomm** account on the NE.
- On the MAE side
 - Select the NE on the MAE-Access topology.
 - Right-click the NE, choose **NE Properties** from the shortcut menu, and change the password of the **emscomm** account on the MAE by specifying **Account for Logging In to NE** in the displayed window.

NetEco

On the NetEco, different passwords of the **emscommneteco** account can be configured for different NEs. To change the password of the **emscommneteco** account, perform the following steps:

- On the NE side
Run the **MOD OP** command to change the password.
- On the NetEco side
Choose **Maintenance > Data Transfer Setting** and change the password in the displayed window of the NetEco.

4.1.1.7 FTP User Management

On GSM and UMTS networks, the BSC6900/BSC6910 has the following FTP users:

- FtpUsr: Uses a third-party FTP client to log in to the FTP server on the NE and then upload or download information about the NE.
- MAE user: Uploads or downloads data between the NE and the MAE.

User management is defined as follows:

- When an FtpUsr user changes the password, the BSC6900/BSC6910 checks the password complexity according to the configured password policy. The BSC6900/BSC6910 does not check the complexity of the password entered by a user during software installation. Instead, the user, when logging in to the FTP server, is prompted with a message indicating that the password complexity is lower than the current configuration and needs to be changed. However, the user can still use the password to log in to the FTP server without interrupting the current FTP connection. The user will be forced to change the password to meet the password complexity requirements when the password expires.
- When an MAE user changes the password, the BSC6900/BSC6910 checks the password complexity according to the configured password policy. However, if an MAE user fails to log in to the FTP server, the BSC6900/BSC6910 does not lock the account but reports a security alarm. This is because the password is used to secure data transmission over the southbound interface between the BSC6900/BSC6910 and the MAE.

In addition, local OM users and domain users also have their FTP rights. On the BSC6900/BSC6910, file transfer and access control rights are listed in the output of the **LST FTPDIRAUTH** command. Operators can customize the rights in the list.

For details, see [Accessing the Web Server Directory Using the BSC6900/BSC6910 File Manager \(GSM/UMTS\)](#).

4.1.2 OMCH Security

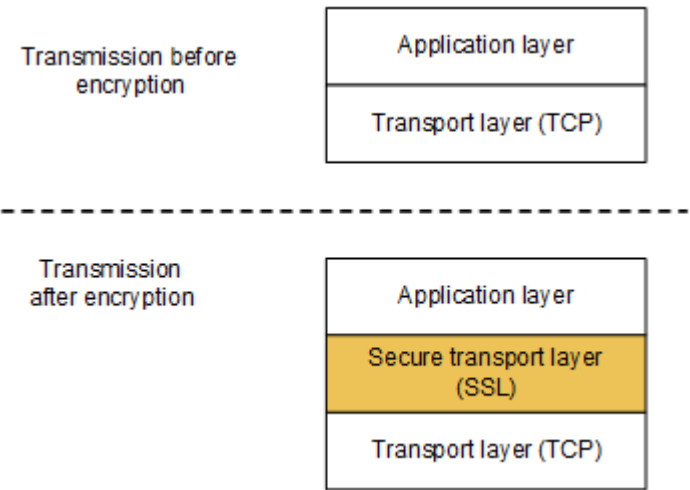
4.1.2.1 SSL-Encrypted Transmission

An OM channel is configured to transmit information about management and maintenance of a base station (other than a GBTS)/base station controller. The channel is established between them and the MAE/WebLMT.

Data transmitted over OM channels is secured using Secure Sockets Layer (SSL).

SSL is a cryptographic protocol that provides communication security over the Internet. As shown in [Figure 4-1](#), SSL works between the transport layer and the application layer to secure data transmission for various application protocols, such as Hypertext Transfer Protocol (HTTP) and File Transfer Protocol (FTP).

Figure 4-1 SSL-encrypted transmission



SSL protects transmitted data against eavesdropping, tampering, and forging using confidentiality protection, integrity protection, and identity authentication.

- Confidentiality protection: SSL encrypts data transmitted between communicating parties to prevent eavesdropping.
- Identity authentication: The communicating parties must authenticate each other before establishing an SSL connection.
- Integrity protection: SSL provides integrity protection for data transmitted between the communicating parties so that the data is not tampered with during transmission.

SSL ensures secure data transmission on the management plane. The OM channel must be protected by SSL. If SSL is disabled for the OM channel, the management plane data may be intercepted or tampered with by attackers, or the NE may even be controlled by attackers. For details about SSL and its application, see *SSL*.

4.1.2.2 SSH-Encrypted Transmission

Secure Shell (SSH) is a network security protocol. It encrypts and authenticates network data to provide secure services, such as access and file transfer, over an insecure network. As a secure alternative to Telnet and other insecure remote shell protocols, SSH has been widely used around the world and supported by most devices.

SSH is supported by wireless base stations in the following scenarios:

- If the MAE is disconnected and both the initial and operator-issued certificates of the UMPT are damaged, the SSH port (6011) is automatically enabled to allow login using the SSH client for local O&M.
- In other scenarios where local O&M through the SSH client is required, the administrator can run the **SET SSHCONF** command to enable the SSH port.

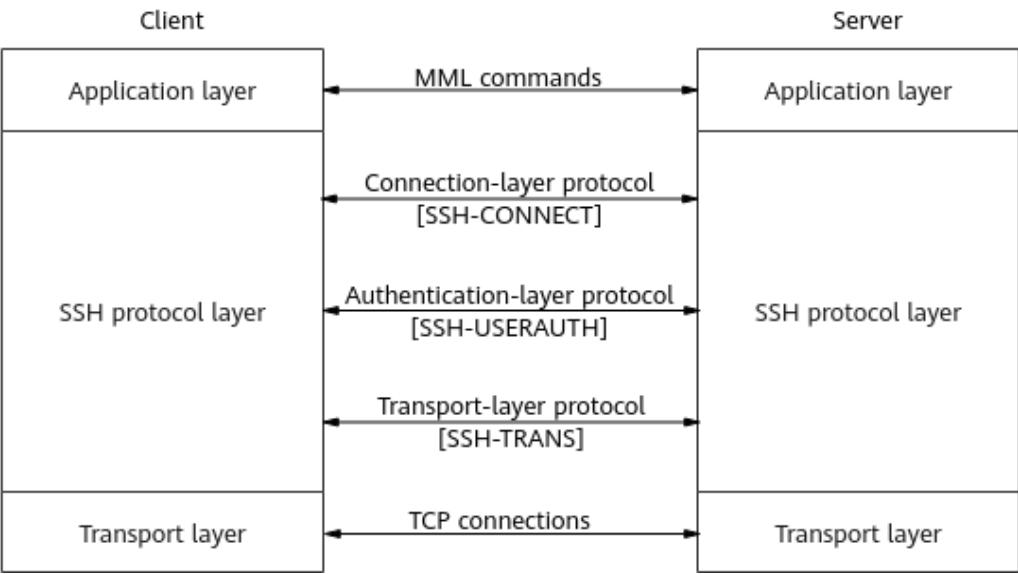
SSH uses encryption and authentication to improve the security of data transmission between two devices. It establishes a secure tunnel between two devices so that the two devices can securely send commands to each other. For example, a user can remotely log in to the SSH server through the SSH client to run commands.

As shown in [Figure 4-2](#), SSH has three protocol layers.

- The transport-layer protocol [SSH-TRANS] runs over TCP connections. It provides server-side authentication, as well as confidentiality and integrity assurance.
- The authentication-layer protocol [SSH-USERAUTH] runs based on the transport-layer protocol. The client uses this protocol to request the server to authenticate a user.
- The connection-layer protocol [SSH-CONNECT] runs based on the authentication-layer protocol. It supports the multiplexing of an encrypted tunnel into multiple logical channels.

SSH-encrypted transmission provides a standard method for secure interactive shell sessions and allows users to run base station MML commands at the application layer.

Figure 4-2 SSH protocol layer structure



NOTE

- SSH does not support the following functions:
- Users who have logged in using SSH are not allowed to modify **Port Enable Flag**.
 - SSH does not support the following commands: **TRACERT**, **TRACERT6**, **PING**, **PING6**, **CFMTRACE**, **CFMPING**, and **UDPECHO**. It provides only basic O&M capabilities. For details about the commands supported by SSH, see *Man-Machine Interface Reference in 3900 & 5900 Series Base Station Product Documentation*.
 - The SSH client does not have a GUI. Therefore, it does not support password modification for the local account or language switching.
- If you need to perform the preceding functions that are not supported, perform related operations on the WebLMT or MAE.

Table 4-4 lists the MML commands and parameters related to SSH-encrypted transmission. For detailed configurations, see 4.4.2.2.2 Data Configuration.

Table 4-4 MML commands and parameters related to SSH-encrypted transmission

MML Command	MML Command Description	Parameter	Parameter Description
ADD/MOD/RMV/ LST SSHPOLICY	Used to add/ modify/remove/ list SSH configuration policies.	SSHPOLICY.POLICYID (LTE eNodeB, 5G gNodeB)	Indicates the SSH configuration policy ID.
		SSHPOLICY.KEYEXCHANGEALG (LTE eNodeB, 5G gNodeB)	Indicates the key exchange algorithm used by SSH sessions.

MML Command	MML Command Description	Parameter	Parameter Description
		SSHPOLICY.CIPHERALG (LTE eNodeB, 5G gNodeB)	Indicates the data symmetric encryption algorithm used by SSH.
		SSHPOLICY.HMACALG (LTE eNodeB, 5G gNodeB)	Indicates the HMAC algorithm used by SSH for data integrity protection.
		SSHPOLICY.HOSTKEYALG (LTE eNodeB, 5G gNodeB)	Indicates the host key algorithm used by SSH for digital signature and authentication.
SET/LST SSHCONF	Used to set/list the SSH configurations.	SSHCONF.POLICYID (LTE eNodeB, 5G gNodeB)	Indicates the SSH configuration policy ID.
		SSHCONF.IDLEFORCESWITCH (LTE eNodeB, 5G gNodeB)	Indicates how to determine whether an SSH session is idle.
		SSHCONF.IDLETIMEOUT (LTE eNodeB, 5G gNodeB)	Indicates the timeout duration of an idle SSH session.
		SSHCONF.MAXLIFETIME (LTE eNodeB, 5G gNodeB)	Indicates the maximum lifetime of an SSH session.
		SSHCONF.PORTFLAG (LTE eNodeB, 5G gNodeB)	Indicates whether to enable the port for NE access over SSH connections.
DSP SSHHOSTKEY	Used to query the SSH host key.	-	-

4.1.2.3 Management-Plane IP Address Isolation

This function isolates the control-plane IP address from the management-plane IP address, preventing users from performing unauthorized operations on the management plane using the control-plane IP address.

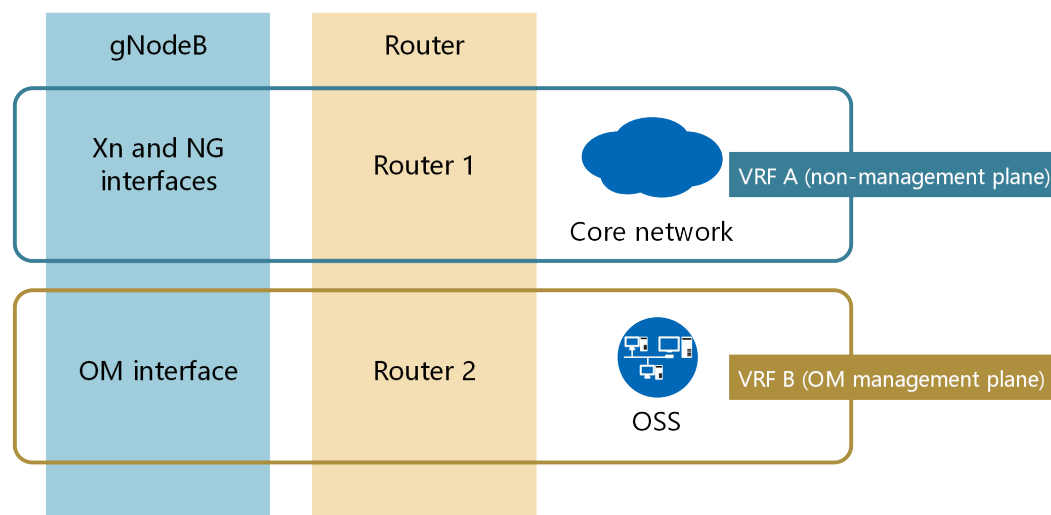
If the **GTRANSPARA.ONLYOMIP (LTE eNodeB, 5G gNodeB)** parameter is set to **ENABLE** and the management-plane IP address is configured, the OM channel between the OSS and the base station must be established using the management-plane IP address. If this parameter is set to **DISABLE**, the OM management plane may be accessed using other IP addresses, increasing the attack surface.

4.1.2.4 Management Plane Isolation by VRFs

The Management Plane Isolation by VRFs function isolates the management-plane IP address from IP addresses of other planes, preventing users from accessing the OM management plane using an IP address of another plane.

To reduce the attack surface of the system, it is recommended that the **OMCH.VRFIDX (LTE eNodeB, 5G gNodeB)** parameter for the OM channel be set on the LTE and NR sides to a value different from that of the VRF configured for non-management-plane interfaces. If this parameter is not specified, the OM channel uses the default VRF, which poses a risk of using non-management-plane IP addresses to access the management plane. [Figure 4-3](#) illustrates the principle using a gNodeB as an example.

Figure 4-3 VRF-based isolation for the management plane of the gNodeB



4.1.2.5 Authentication between the EMS and NEs

Challenge-response authentication is used to ensure user login security. In challenge-response authentication mode, each time the authentication server sends a different question ("challenge") to the client, which must provide a valid answer ("response"). Authentication is implemented by exchanging the digest value of the random number and password, instead of simply transmitting passwords. The challenge-response authentication mechanism protects passwords against disclosure and replay attacks.

4.1.2.6 OMU Anti-Attack for the BSC6900/BSC6910

The integrated firewall performs the following operations on all IP data streams transmitted to the OMU:

- IP address filtering
The OMU accepts IP data streams only from authorized IP addresses and network segments.
- Defense against attacks, such as ICMP ping, IP fragmentation, low time to live (TTL), Smurf, and distributed denial-of-service (DDoS) attacks
- Defense against TCP sequence prediction attacks and synchronization (SYN) flood attacks
- Isolation between internal and external networks for the BSC6900/BSC6910 on GSM and UMTS networks
Packets whose destination IP addresses are internal IP addresses or belong to an internal network segment cannot enter the BSC6900/BSC6910 through the OMU.

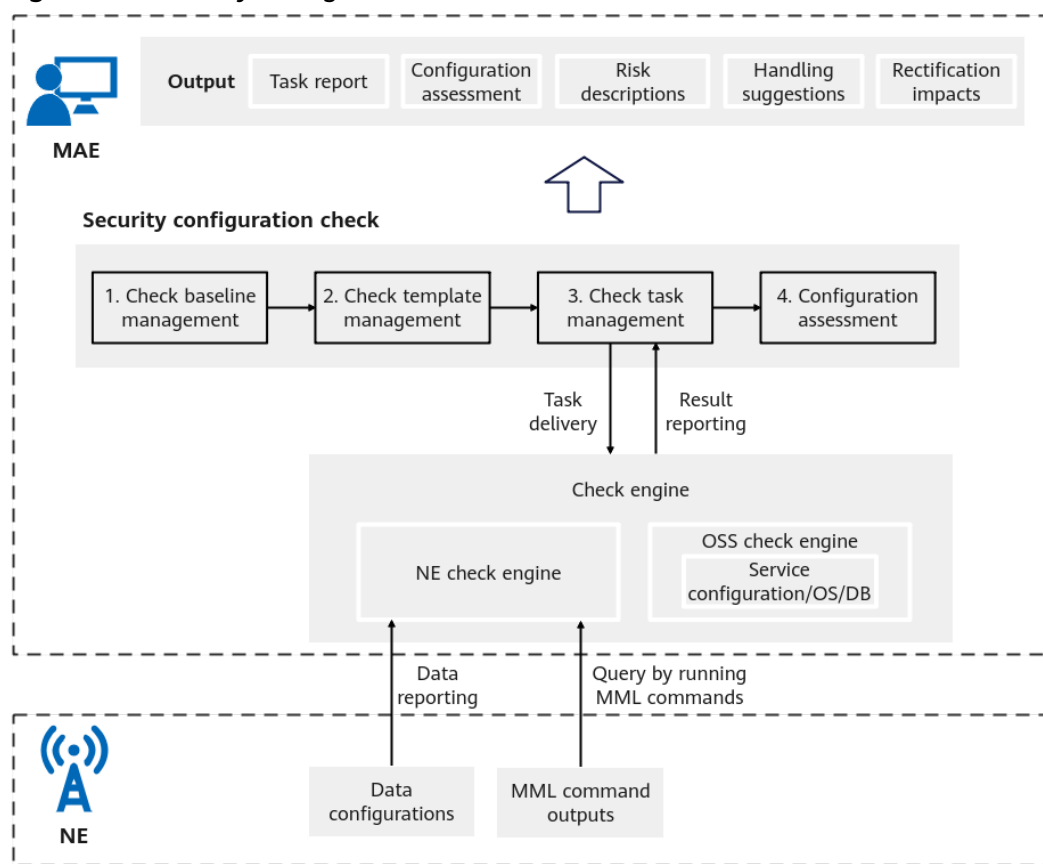
For a properly running network, specifying whitelisted and blacklisted IP addresses is generally not required and IP addresses used for access are not restricted. Specifying whitelisted and blacklisted IP addresses improves the security of the BSC6900/BSC6910 on GSM and UMTS networks:

- Whitelist: Only the specified IP address or IP addresses in the specified network segment can be used to access the BSC6900/BSC6910. The whitelist can be configured for a particular port or for all ports. If some IP addresses or network segments are whitelisted, other IP addresses or network segments are blacklisted and cannot be used to access the BSC6900/BSC6910.
- Blacklist: The specified IP address or IP addresses in the specified network segment cannot be used to access the BSC6900/BSC6910. The blacklist can be configured for a particular port or for all ports. All IP addresses that are not blacklisted are actually in the whitelist.

4.1.3 Security Configuration Check

4.1.3.1 Overview

The security configuration check function checks security-related parameter configurations of an NE according to specified check rules and provides risk descriptions, handling suggestions, and rectification impacts in the check results. OM personnel can then query the check results, perform comprehensive evaluation, determine the parameter reconfiguration policy, and perform reconfiguration to mitigate the device security risks. Security configuration check involves check baseline management, check template management, check task management, and configuration assessment, as shown in [Figure 4-4](#).

Figure 4-4 Security configuration check

Users can perform security configuration check on live-network data by choosing **Access > Security > Security Configuration Management > Security Configuration Check** on the MAE GUI. In addition, the system provides a default check task to periodically check live-network data. After a check is complete, the risk level is displayed. Based on different check scenarios and risk notification requirements, OM personnel can perform the following operations:

- Adjust the expected values, risk levels, and inapplicable versions of the check rules.
- Configure the risk levels displayed as required.

For details, see sections related to security configuration check in MAE product documentation.

NOTE

NEs involved in this function refer to the base station and BSC6900/BSC6910.

From OSS21.1 onwards, the security policy level function of the base station is independent of the original consistency check feature.

From OSS22.1 onwards, the security policy level function of the BSC6900/BSC6910 is moved from the consistency check feature to the security configuration check feature.

From OSS23.1 onwards, the security configuration check function on the MAE-Deployment is moved to the MAE-Access.

4.1.3.2 Check Baseline Management

Check baseline management allows OM personnel to adjust the default values of, import, export, and categorize the check rules provided by the system. The default values of the check rules include expected value, risk level, inapplicable version, and rule category. OM personnel can manage the check rules as follows:

- Change the default values of the check rules to the values that comply with the security policies of the live network or restore the default values.
- Export check rule files and import the files to other MAE systems.
- Customize security categories and security requirements of the check rules.

Check rules are categorized as follows:

- Common security policies, such as password policies
- Security policies aiming to reduce the attack surface, such as policies for local maintenance ports
- Security policies for protocols and algorithms, such as policies for TLS

NOTE

- The check rules in check baseline management are for check templates. Therefore, modifications to these check rules only affect check templates created after the modifications and do not affect existing check templates.
- The default security categories cannot be modified or deleted.
- A maximum of 50 security categories and 150 security requirements can be customized for baselines of different products. Baselines of different products (such as the base station and BSC6900/BSC6910) are independent of each other.

4.1.3.3 Check Template Management

Check template management is used to manage check rules by groups. This helps specify the check rule range when a check task is created. The system provides two default check templates. The "Default check template" involves check rules for ensuring basic security of the management plane, control plane, and user plane of a base station. Such rules include whether to use insecure algorithms and protocols and whether to enable encrypted channels on the control plane and user plane. The default periodic task is performed based on the check rules in the "Default check template". The other default template is named as "BTS GSMA FS.31 3.0 Controls", which complies with the security requirements of FS.31 GSMA Baseline Security Controls 3.0 and can be used if required.

Operators can also create and delete check templates based on their own security baselines and set different expected values and inapplicable versions for different security check scenarios. The created check templates can be exported from and imported into different MAE systems.

 NOTE

- Check templates can only be exported from and imported into MAE systems of the same language.
- It is recommended that check templates be exported from and imported into MAE systems of the same version. If the check template of a different version is imported, issues such as rule verification failures and file format mismatch may occur. In this case, you need to create a check template again.
- You can modify data such as the expected values of check rules in check templates. The modification in one template does not affect the same check rule in other templates.

4.1.3.4 Check Task Management

Check task management allows OM personnel to create, delete, modify, and execute check tasks. During check task creation, check templates are used to specify the check rules and NEs to be checked are selected as required. A maximum of 20 check tasks can be created.

Check tasks include real-time tasks and periodic tasks. Real-time tasks need to be manually executed, while periodic tasks are executed at the specified interval. Periodic tasks can be suspended when not required. Only one task can run at a time.

After the check is complete, OM personnel can view the check report to identify the parameter configurations that do not satisfy the check rules, information about the NEs that do not satisfy the check rules, as well as the risk descriptions, handling suggestions, and rectifications impacts of the rules that are not satisfied.

4.1.3.5 Configuration Assessment

Configuration assessment scores the configurations in both asset view and rule view based on the results of all completed check tasks. In addition, it categorizes and collects statistics on the rules that are not satisfied, and analyzes the distribution of the rules that are not satisfied.

The score is calculated as follows:

- The score of a check rule is calculated by the system by default based on factors such as the risk level, impact on services, and application scenario. Each check rule has a unique weight score.
- The score of a check task is calculated as follows: $\text{Score} = \frac{\text{Score of satisfied rules}}{\text{Target total score}}$
 - $\text{Score of satisfied rules} = \text{Total score of rules that are satisfied in the check task}$
 - $\text{Target total score} = \text{Total score of rules that are satisfied and those that are not satisfied in the check task}$

A high target total score indicates that a large number of rules are selected and the check task covers a wide scope. Alternatively, it means that the selected check rules have high risk levels. When they are not satisfied, services are significantly affected.

A high score of satisfied rules indicates that a large number of rules are satisfied in the check task, or the rules with high risk levels and great impact on services are satisfied.

4.1.4 Application Security

4.1.4.1 Web Security

4.1.4.1.1 Overview

By logging in to the WebLMT, users perform OM on the base station/base station controller. The WebLMT is an HTTP/HTTPS-based web application that takes the following measures to ensure OM security:

- HTTPS-based data transmission
- Anti-attack

4.1.4.1.2 HTTPS-based Data Transmission

By default, the WebLMT uses HTTPS to secure data transmission. When HTTPS is required, digital certificates must be used. In this case, digital certificates issued by the operator's CA system must be configured on the NE. For details about digital certificate management, see *PKI*. The value **HTTPS_ONLY** forcibly takes effect as the WebLMT login policy.

Table 4-5 WebLMT login policy

Policy	Protocol Used in the Browser Address Bar	Protocol Used in Login Web Page	Protocol Used in the WebLMT GUI
HTTPS_ONLY	HTTP	HTTPS	HTTPS
	HTTPS	HTTPS	HTTPS

4.1.4.1.3 Session Management

The WebLMT server supports the session idle timeout and session lifetime management to control the session duration.

1. Session Idle Timeout

In the idle state, no user operation is performed. A session will time out if it remains in the idle state for a specified period. When the idle period of a session reaches the configured timeout duration, the session will be cleared.

The **WEBLMT.IDLETIMEOUT (5G gNodeB, LTE eNodeB)** parameter specifies the timeout duration of session idleness.

A tracing task, monitoring task, or batch processing task of a base station may last for hours, and users can hardly stay active for such a long time. To prevent UE tracing and monitoring tasks from being interrupted due to idle session timeout, a switch for forcing a session to enter the idle state upon expiration has been added. The **WEBLMT.IDLEFORCESW (5G gNodeB, LTE eNodeB)** parameter specifies whether to force a session without user operations to enter the idle state. If this parameter is set to **OFF**, a session will not be considered idle when a tracing and monitoring task is ongoing. If this

parameter is set to **ON**, a session will be considered as idle and enter the idle state even when a tracing and monitoring task is ongoing. In this case, the tracing and monitoring task is forcibly stopped.

It is recommended that the **WEBLMT.IDLETIMEOUT** parameter be set to a value indicating 10 minutes.

It is recommended that the **WEBLMT.IDLEFORCESW** parameter be set to **ON**. If this parameter is set to **OFF** and a user performs a tracing or monitoring task for a long period of time, the task is still valid even if no user operation is performed. In this case, information may be disclosed or attackers may perform malicious operations. If this parameter must be set to **OFF**, you are advised to lock the screen of the PC where the WebLMT client is installed before leaving.

2. Session Maximum Lifetime

The maximum lifetime of a session refers to the maximum duration that a session can survive. A session is cleared when its maximum lifetime expires, regardless of whether any operation or task is being performed.

The **WEBLMT.MAXLIFETIME** parameter specifies the maximum lifetime of a session.

You are advised to set the **WEBLMT.MAXLIFETIME** parameter to **480** (in minutes), indicating eight hours. The smaller the value of this parameter, the shorter the maximum lifetime of a session. If this parameter is set to an excessively small value, sessions will be cleared in a short period of time, interrupting user tasks. The larger the value of this parameter, the longer the maximum lifetime of a session. An excessively large lifetime of a session increases the risk of session interception. Therefore, you are not advised to set this parameter to a value greater than 480 (in minutes).

In addition, when the administrator resets the password or permission of a local user, all sessions of the user are automatically terminated. When a local user or domain user changes its password through a session, other sessions of the user are automatically terminated.

4.1.4.1.4 Anti-attack

The web server has been reinforced to prevent the impacts of various attacks. The following types of attacks have been considered before delivery:

- Cross-site scripting attack
Attackers inject malicious scripts into web pages. If the web server does not filter out the malicious scripts, the scripts will be executed when users view the web pages.
- Remote file inclusion attack
Attackers forcibly include their files in the codes on the web server by exploiting the web server's vulnerability in filtering file inclusion. By doing this, the attackers can attack certain websites.
- Directory traversal attack
Attackers use the security holes of applications to access data or directories without obtaining authorization, thereby causing data leak or tampering.
- Distributed denial of service (DDoS) attack
Attackers use the inherent security holes of network protocols to forge reasonable requests to consume limited transmission bandwidth or occupy

excess resources. As a result, the network or service cannot properly respond to authorized requests and breaks down.

- Structured Query Language (SQL) injection attack

SQL injection attacks are a common type of injection attacks. Attackers inject malicious SQL commands into a web form entry to trick the web server into executing the SQL commands.

- Broken authentication and session management attack

Attackers exploit the defects in functions related to identity authentication in web applications to steal authentication information or session management data, causing user or administrator account thefts.

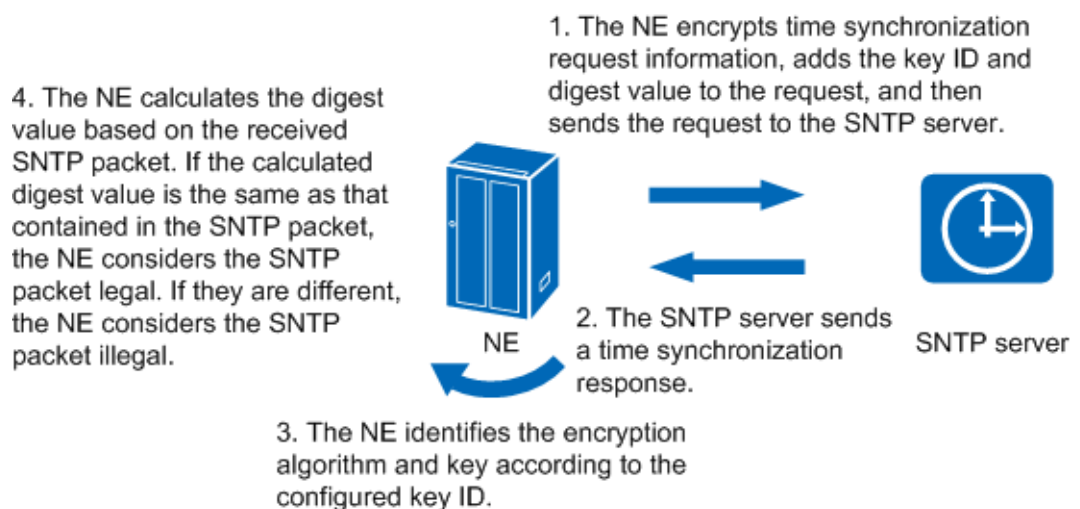
4.1.4.2 Time Security

4.1.4.2.1 SNTP Security for the BSC6900/BSC6910 (GSM/UMTS)

An NE must synchronize its time with the Simple Network Time Protocol (SNTP) server (for example, the MAE) to ensure that the system time is accurate. Time synchronization uses SNTP and supports plaintext mode or authentication mode, which is specified by the **SNTPSRVINFO.AUTHMODE** parameter. The authentication mode ensures SNTP security.

SNTP security prevents the NE from incorrectly adjusting the time after receiving a time synchronization attack message. This ensures the security of the NE on the network and guarantees normal OM functions. [Figure 4-5](#) shows an SNTP time synchronization process.

Figure 4-5 SNTP time synchronization process



The NE currently supports the SNTP V3 protocol and is compatible with the SNTP server and NTP server. However, the time synchronization precision of the NE is the same as that supported by the SNTP server.

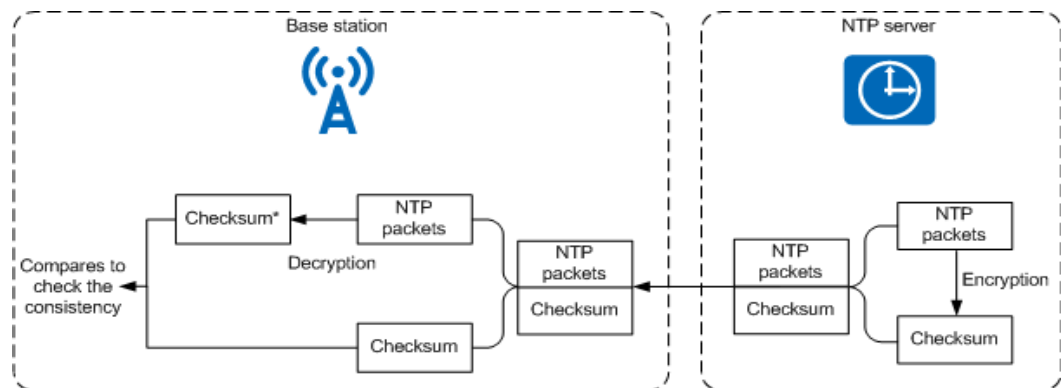
4.1.4.2.2 NTP Security Authentication for the Base Station/BSC6960

A base station/BSC6960 deployed on a public network may obtain incorrect time from an invalid reference clock, resulting in inconsistent time between NEs. This

will generate erroneous time information in records, such as in alarm information and logs, affecting the maintainability.

NTP security authentication protects the integrity of and authenticates the source of NTP packets received by the base station/BSC6960 to ensure that it uses valid reference clocks. The **NTPCP.AUTHMODE (5G gNodeB, LTE eNodeB)**, **NTPCP.KEY (5G gNodeB, LTE eNodeB)**, and **NTPCP.KEYID (5G gNodeB, LTE eNodeB)** parameters on an NTP client must be set to the same values as those on the NTP server. NTP security authentication supports the encryption algorithms of Data Encryption Standard (DES), message-digest algorithm 5 (MD5), and SHA256. The DES and MD5 encryption algorithms are not recommended because they have been cracked and will be deleted in later versions. NTP security authentication uses digital signatures to verify NTP packets to ensure the validity of the reference time received by the base station/BSC6960. Figure 4-6 illustrates the principle for NTP security authentication for a base station.

Figure 4-6 Principle for NTP security authentication



If the **NTPCP.AUTHMODE (5G gNodeB, LTE eNodeB)** parameter is not set to **PLAIN**, NTP security authentication is performed in encryption mode.

If the **NTPCP.AUTHMODE (5G gNodeB, LTE eNodeB)** parameter is set to **PLAIN**, the NTP server sends unencrypted NTP packets to the base station/BSC6960. Therefore, the base station/BSC6960 does not need to decrypt the received NTP packets.

4.1.5 Security Monitoring

4.1.5.1 Security Monitoring

Immediate or periodic monitoring tasks are performed to monitor external access, account lists, software versions, and system running process lists.

According to the preceding information on the NE, security administrators of operators can determine whether an NE has been attacked, take protective measures in a timely manner, and reduce risks.

Users can create a one-time or periodic monitoring task by performing the following operations on the MAE-Access: Choose **Security > Security Posture > Security Status Monitoring**, select **External access monitoring**, **Account list monitoring**, **Software patch installation monitoring**, or **system running process monitoring** under **Monitoring Type**, and select NEs.

Software Version Blacklist Monitoring

The software version blacklist monitoring function enables users to add a version of an NE to the software version blacklist on the MAE if major security vulnerabilities are found in the version. The MAE periodically checks software versions of NEs on the live network. If an NE software version has been blacklisted, the MAE generates a critical alarm.

Operators need to maintain a software version blacklist on the MAE. If an NE version has been blacklisted, the MAE prevents users from uploading the software package of this version to the MAE server. Software packages that have been loaded to the MAE server are not affected.

This function is performed only on the MAE.

4.1.5.2 Security Alarms, Events, and Logs

4.1.5.2.1 Overview

The MAE and WebLMT manage security alarms, events, and logs. If security faults occur, users can be informed of the faults and perform fault diagnosis according to the reported alarm or event information. In addition, security risks and vulnerability can be analyzed by tracing history security alarms and logs. Detailed information about the traced objects is recorded in the tracing logs.

4.1.5.2.2 Security Alarms and Events

Table 4-6 lists the security alarms and events that may be reported by the BSC6900/BSC6910 when the related security faults occur on GSM and UMTS networks.

Table 4-6 Security alarms and events

Alarm or Event ID	Alarm or Event Name
ALM-20723	File Loss or Damage
EVT-22813	Domain User Login Failed
EVT-22814	Local User Login Failed
EVT-22815	Local User Locked
EVT-22805	Local User Modifying Other Operator's Password
ALM-20732	SSL Certificate File Abnormality
ALM-20850	Digital Certificate Will Be out of Valid Time
ALM-20851	Digital Certificate Loss, Expiry, or Damage
ALM-20852	Exceeded Failures of Logins by the Local User
ALM-20714	OMU Time Synchronization Abnormity

Table 4-7 lists the security alarms and events that may be reported by the BSC6960 when the related security faults occur.

Table 4-7 Security alarms and events

Alarm or Event ID	Alarm or Event Name
ALM-20973	Board Not In Position
ALM-20957	Water Alarm
ALM-20958	Smoke Alarm
ALM-20959	Burglar Alarm
ALM-21028	Local User Consecutive Login Retries Failed
ALM-21665	Invalid Packet Exceeded
ALM-20991	Time Synchronization Failure
ALM-20976	Board File System Damaged
EVT-21043	User Login to the NE
ALM-21028	Local User Consecutive Login Retries Failed
ALM-21029	Peer Certificate Invalid
ALM-21030	Imminent Certificate Expiry
ALM-21031	Certificate Invalid
ALM-21032	Automatic Certificate Update Failed

Table 4-8 lists the security alarms and events that may be reported by the base station when the related security faults occur.

Table 4-8 Security alarms and events

Alarm or Event ID	Alarm or Event Name
ALM-26204	Board Not In Position
ALM-25670	Water Alarm
ALM-25671	Smoke Alarm
ALM-25672	Burglar Alarm
ALM-26830	Local User Consecutive Login Retries Failed
ALM-25950	Base Station Being Attacked
ALM-26266	Time Synchronization Failure

4.1.5.2.3 Security Logs and Security Audit

The base station/base station controller supports the log management function. This function records security operations and events during routine O&M and prohibits modification of records. Based on the recorded information, the operators can perform security audit, identify sources of security accidents and problems, and find ways to improve network security.

Logs record information about system security and user operations, and are classified into operation logs, system logs, and security logs of NEs and the MAE. By querying logs, users can obtain information about the running status of, system security situation of, and user operations on NEs or the MAE. Users can also save logs as files or print them out.

The MAE can centrally manage NE logs as follows:

- Centrally collects, queries, measures, analyzes, and outputs logs.
- Records log information about its own running status, security events, and operations, which is used for query and audit.
- Periodically collects NE logs based on user settings

Users can audit the security logs collected by the MAE to evaluate OM security.

OM Event Recording

Logs of the MAE, base station controller, and base station independently record information about system security and user operations, that is, OM security-related events during the running process.

Operation Logs

Operation logs on the MAE and NEs record operations automatically triggered by the system as well as maintenance operations manually performed on the WebLMT or MAE. Manual operations include deletion, modification, creation, query, loading, and switchover.

Operations logged are classified into the following types:

- MML command execution, for example, execution of a command to add a user:
Operation="ADD OP: OP="user",..."
- Automatically triggered intra- or inter-system operations. These operations may not be directly performed by users, for example, addition of a measurement object:
Operation="ADD MEASOBJBYID: MOC=50331652,..."
- Combined operations for an action on the GUI, for example, operations for creating a tracing task:
Operation="CmdCode=0x00000101 CmdName="Create Trace Task" TaskID=4294963333..."

NOTE

For operations that cannot be performed directly by users, contact Huawei technical support to obtain details about the operation logs.

System Logs

System logs mainly record the system running status of NEs or the MAE. System logs help users to learn about the system running status, fault diagnosis, and

location progress and status of security accidents. The system herein refers only to Huawei-developed application systems. The system logs include those of the MAE and NEs.

System logs record the following information:

- Abnormal status and actions while the system is running, such as active/standby switchovers, storage failures, and timer expiration
- Key events during system running, such as system startup and shutdown
- Operating status of the system process, such as the process start, exit, running, and abnormality (for example, the system process stops responding)
- Usage of system resources, such as central processing unit (CPU), memory, and hard disk

Security Logs

Security logs record information about security events.

Security logs of base stations record the following:

- Events related to account login, such as user login, user logout, account locking, and account unlocking
- Events related to account management, such as account addition, deletion, and modification, password change, and permission modification
- Events related to user authentication, such as unauthorized access
- Events related to network exceptions, such as reception of invalid packets

The security logs include those of the MAE and NEs. Users can evaluate system security by auditing security logs. For details, see [Security Log Auditing](#).

[Table 4-9](#) describes security events recorded in security logs that the BSC6900/BSC6910 can provide on GSM and UMTS networks.

Table 4-9 Security logs of the BSC6900/BSC6910

Security Event Type	Security Log
Account login event	A domain user has logged in to the NE.
	A domain user has logged out of the NE.
	A local user has logged in to the NE.
	A local user has logged out of the NE.
	The system locks a local user account whose failed login attempts exceed the maximum number.
	The system automatically unlocks a local user account after the locking time expires.
	A local user account is manually unlocked.

Security Event Type	Security Log
	A local user account is locked by the administrator.
	An account is automatically locked when the password expires.
Account management event	A domain user or local user has been forced to log out after having logged in to the NE.
	A local user account has been added, removed, or modified.
	The user group to which a local user belongs has been changed.
	The rights granted to a local user group have been changed.
	The commands in a command group have been adjusted.
	The rights granted to a local user have been changed.
	A local user has changed the user's password.
	A local user has changed the password of another user.
	The account or password policy has been changed.
OMU security event	The OMU has started or stopped, or active and standby OMUs have been switched over.
Digital certificate security event	A digital certificate has been updated.
Upgrade-related security event	The driver has been upgraded.
OMU configuration-related security event	OMU network parameters, such as the internal network, external network, VLAN, mask, IP address, and host name, have been modified.
	Active and standby OMUs have been configured.
OMU security event for changing the password of an initial privileged account	The password of the administrator account has been changed.
	The password of a database account has been changed.
SNTP time synchronization event	SNTP time synchronization has failed.

Table 4-10 lists security-related operation logs that the BSC6900/BSC6910 can provide on GSM and UMTS networks.

Table 4-10 Security-related operation logs

Security Event Type	Operation Log
Account authentication events	A domain user or local user fails to be authenticated to perform a certain operation.
	A user attempts to access an object without the permission, which is specified when the user is created by running the ADD OP command.

The **LST SECLOG** and **LST OPTLOG** commands can be used to query security logs and operation logs, respectively.

Centralized Log Management

The MAE supports the following centralized management of MAE logs and NE logs:

- Log collection
The MAE can periodically collect NE logs based on user settings. Users can also set dumping and export of MAE logs and NE logs.
- Log query and printing
Users can obtain information about the running status of, system security situation of, and user operations on NEs or the MAE. Users can also save logs as files or print them out.
- Log analysis
Based on the collected MAE logs and NE logs, users can analyze information such as system running status, security events, and operations.

Log Collection

Users can collect and dump all operation logs, security logs, and system logs of the MAE as well as operation logs and security logs of NEs. NEs generate and save their own system logs and automatically report the logs to the MAE. For details, see the "Log Management" section in *MAE Product Documentation*.

Log Query and Printing

For details about how to query or print logs on the MAE, see the "Log Management" section in *MAE Product Documentation*.

On the WebLMT, users can query log files generated during a specified time range, including operation logs and security logs. For details about how to query the logs, see *3900 & 5900 Series Base Station MML Command Reference* in *3900 & 5900 Series Base Station Product Documentation*.

Security Log Auditing

Auditing Security Events

Security event auditing refers to a process in which the base station/base station controller generates audit records based on security events (security logs). Auditable security events include:

- Startup and shutdown of the system or applications
- User login success and failure events: Including information about user names, login time, workstation (such as its IP addresses), and causes of login failures (such as incorrect passwords and invalid accounts)
- User logout success and failure events: Including information about user names, logout time, workstation (such as its IP addresses), and causes of logout failures
- Users' attempt to access resources without their permission
- All O&M and configuration events: Including information about user names, O&M time, workstation (such as its IP addresses), operations, and responses
- Operations concerning user accounts and permission levels: Including addition, deletion, and modification

Events to be recorded in security logs are configurable, and the configuration process must be recorded as security events that can be audited.

For details about how to audit security logs, see the "Log Management" section in MAE product documentation.

Saving Security Logs

The base station/base station controller uses databases to save security logs. Users cannot modify or delete these logs.

If the number of audit records saved in any security log exceeds 200,000, the base station/base station controller transfers the earliest 10,000 records to a flash memory to prevent the database from overflowing.

If the number of saved logs reaches a limit, earliest logs will be discarded at the arrival of new logs.

Run the **SET LOGLIMIT** command to configure the maximum number of logs that can be saved on the BSC6900/BSC6910. This number cannot be configured on the base station.

Querying Security Logs

Users can query available audit records in databases. The base station/base station controller supports query by time interval, user name, interface, workstation IP address, result, and command name (for example, MML command names).

For details about how to query security logs, see the "Log Management" section in MAE product documentation.

Log Management Policies

Encryption of Locally Stored Operation Logs and Security Logs

eNodeBs, gNodeBs, and co-MPT base stations protect the confidentiality of locally stored operation logs and security logs using symmetric encryption algorithms. This protection mechanism only enhances the internal security capability of the base stations and does not affect the existing log query and export functions.

4.1.5.2.4 NE Resource Monitoring

Users can run the **DSP PROCESSINFO** command to query the process information of the base station and base station controller. The process information contains the process name, process ID, CPU usage, memory usage, start time, and process description.

4.1.6 Data Security

4.1.6.1 Personal Data Security

4.1.6.1.1 User Identity Security Processing

- GSM and UMTS
To protect personal privacy, Huawei equipment supports user data pseudonymization. This function generates pseudonyms to replace user identity information involved in OM. For details about how to enable this function, see *User Data Pseudonymization* in GBSS feature documentation or RAN feature documentation.
- LTE
 - When video experience optimization is enabled, the eNodeB replaces UE IP addresses with pseudonyms in the reported information to protect personal privacy in the following scenarios:
 - The SC algorithm switch is turned on, and single-UE signaling tracing on the entire network is enabled.
 - The external CHR event PERIOD_UE_SVCTCP_PLAY_SC has been subscribed to.

NOTE

Video experience optimization is enabled by maintenance personnel of operators. To configure a pseudonymity key on the MAE-Access, choose **Monitor > Pseudonymization Policy Management** (under **Signaling Trace**). The **Pseudonymity Key Management** tab page is displayed.

- The **STMSI_MASK_SW** option of the **eNodeBChrOutputCtrl.ChrOutputCtrlSwitch** parameter needs to be selected on the eNodeB to allow the eNodeB to replace the value of the S-TMSI field with all 0s in generated CHR for TMSI protection.
- In other scenarios, the eNodeB anonymizes personal data (such as the IMSI as well as payload in user-plane packets) in OM data by default. This function takes effect by default and is not controlled by a switch.

- NR
 - The **STMSI_MASK_SW** option of the **gNodeBParam.ChrSwitch** parameter needs to be selected on the gNodeB so that the gNodeB does not include the S-TMSI in generated CHR for TMSI protection.
 - In other scenarios, the gNodeB anonymizes personal data (such as the IMSI as well as payload in user-plane packets) in OM data by default. This function takes effect by default and is not controlled by a switch.

4.1.6.1.2 Personal Data Protection

To protect personal data, Huawei supports the following:

- Specifying and logging the causes for starting system tasks that involve personal data. The system tasks mainly include trace tasks.
- Periodically deleting files that contain personal data on the BSC6900/BSC6910. Operations include:
 - Set the **USEREVTRTNPOLICY.TraceDelPeriod** parameter to specify the interval of deleting local tracing files.
 - Set the **USEREVTRTNPOLICY.CallLogDelPeriod** parameter to specify the interval of deleting local CHR and MR files.
 - Set the **USEREVTRTNPOLICY.MBSCLogDelPeriod** parameter to specify the interval of deleting other local files.
- Periodically deleting files containing personal data that has been stored for more than 28 days on base stations.

4.1.6.2 Security Management of Configuration Files

4.1.6.2.1 Overview

The configuration data contains some security-sensitive data, such as keys and passwords. The security-sensitive data is encrypted when being stored in the system database. When the configuration data is exported to a configuration file, the configuration file can be encrypted by adding a password. If the exported configuration data does not need to be imported back to the base station and sensitive data (such as keys and passwords) is not required, the sensitive information can be anonymized in the configuration file.

If the configuration data is not encrypted or anonymized when being exported to a configuration file, the configuration file may contain security-sensitive fields. In this case, the operator must store the configuration file properly and delete it in time to avoid information leakage.

4.1.6.2.2 Application Scenarios

Configuration file encryption applies to the following scenarios:

- Offline transmission of a configuration file
 - The configuration scripts are exported from the MAE-Deployment and then copied to an NE for activation.
 - The configuration scripts are exported from an NE and then copied to another NE for activation.

- Permanent storage of configuration files
NE data (including scheduled tasks) is backed up online on the MAE.

Configuration file anonymization mainly applies when the configuration data of the base station needs to be exported from NEs into configuration scripts and imported into various analysis tools.

4.1.6.2.3 Configuration File Encryption

The **ENCRYPTMODE** parameter specifies the encryption mode and it has three values:

- **UNENCRYPTED**
- **PWD_ENCRYPTED**. In this case, a password consisting of 6 to 64 characters is required.
- **ANONYMIZED**

Figure 4-7 shows the procedure for transmitting an encrypted configuration file in offline mode.

Figure 4-7 Offline transmission of a configuration file

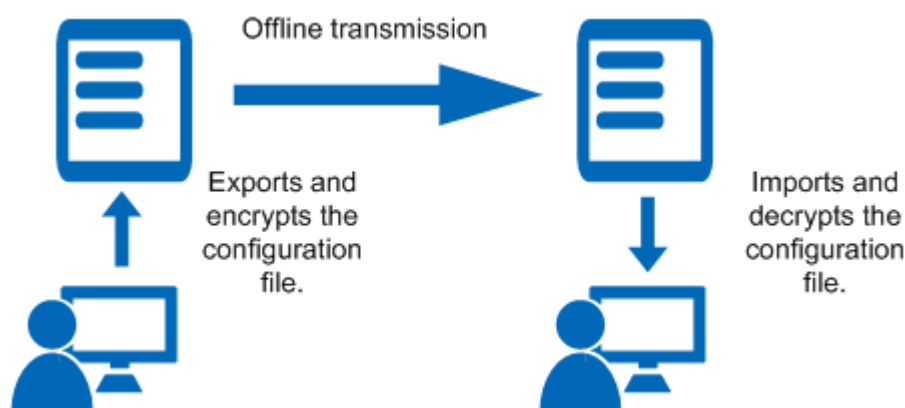
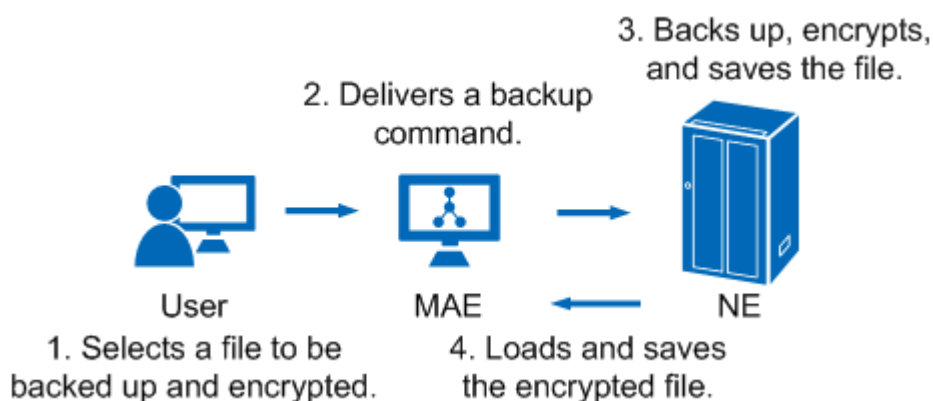


Figure 4-8 shows the procedure for storing a configuration file permanently in online mode, with online backup of NE data on the MAE as an example.

Figure 4-8 Online permanent storage of the configuration file



The following changes have been added to support configuration file encryption:

- The **ENCRYPTMODE** and **FILEPWD** parameters are added to the southbound interface commands and MML commands.
- Encryption and decryption options are added to the GUI of the MAE, Web LMT, upgrade tools, and other tools.

4.1.6.3 Digital Signature-based Software Integrity Protection

4.1.6.3.1 Definition

Software integrity protection adds a digital signature to software by using a private key before uploading software to the target server or NE. When a target NE downloads, loads, or runs software, the NE authenticates the digital signature by using a matched public key. This ensures end-to-end software reliability and integrity.

With this function, any virus or software tampering can be promptly detected. This prevents malicious software from running on NEs.

4.1.6.3.2 Application Scenarios

Software integrity protection applies to the following scenarios:

- Software installation
- Software upgrade
- OS (DOPRA Linux or Euler Linux) upgrade
- OS (DOPRA Linux or Euler Linux) driver upgrade

4.1.6.3.3 Software Digital Signature

Overview

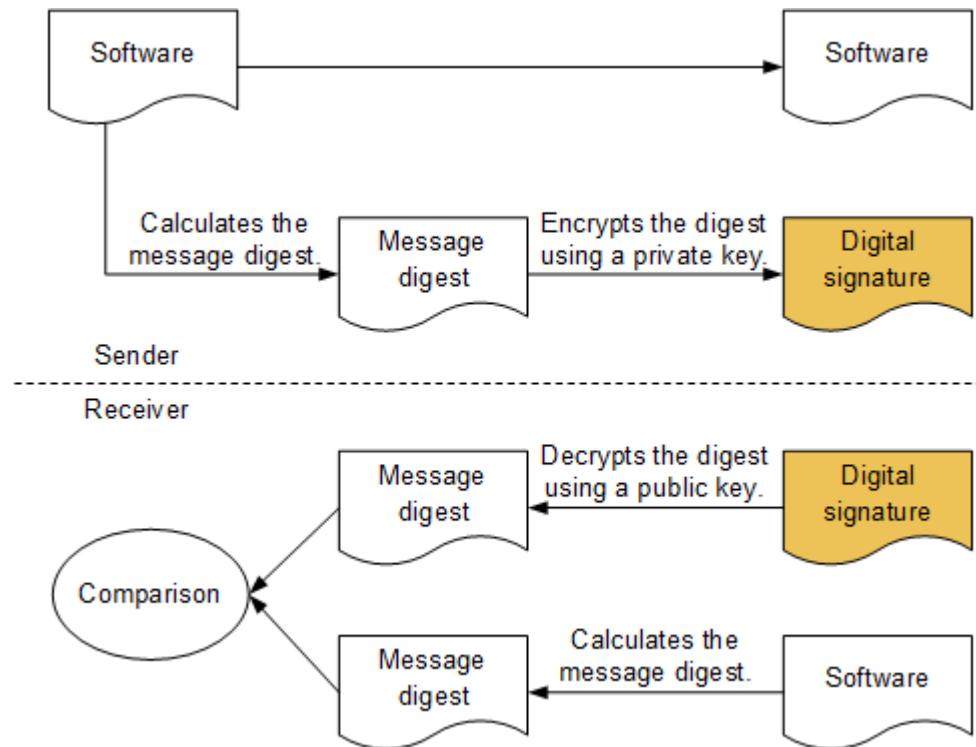
Integrity protection uses digital signatures with the following techniques adopted:

- Hash algorithm: A one-way hash function that converts a data block of an arbitrary length into a fixed-length bit string. Hash algorithms are used as digital signature digest algorithms in this document.
- Rivest-Shamir-Adleman (RSA) public key cryptography: A pair of public and private keys that are used for encryption and decryption. The two keys belong to the same holder and must match each other. The public key can be publicly used, whereas the private key must be kept confidential. RSA algorithms are used as digital signature algorithms in this document.

Principles

Figure 4-9 illustrates the principles of software digital signature.

Figure 4-9 Software digital signature principles



The procedure for adding a software digital signature is as follows:

1. A hash algorithm calculates the message digest for a file to be signed in the software package.
2. The private key is used to encrypt the message digest.
3. The encrypted message digest is saved to a digitally signed file.

The digitally signed file and original software package are then released as the new software package.

After an NE or the MAE receives the new software package, it verifies the digital signature. The procedure for verifying the software digital signature is as follows:

1. The same hash algorithm as that in the preceding procedure calculates the message digest for the file to be verified in the software package.
2. The public key is used to decrypt the digitally signed file to restore the message digest.
3. The restored message digest is compared with the message digest that is calculated using the hash algorithm.

If they are identical, the software is not tampered with. If they are different, it is determined that the software has been tampered with.

PKI-CMS Digital Signature Solution

The PKI-CMS digital signature solution uses Huawei digital signature server and Huawei root certificate to verify signature certificates. In this way, digital signatures are audited and private keys are securely stored, preventing signature abuse and private key disclosure.

The SHA256 verification algorithm and a 4096-bit RSA private key are used to generate digital signatures. The private key is stored on Huawei digital signature server. Huawei digital signature server uses the private key to generate two digital signatures. The digital signatures generated from verification codes are used to verify software integrity. The digital signatures generated from software packages are used to check whether the software packages are released by Huawei.

During digital signature verification, the MAE, upgrade tool, or NE uses the root certificate in the CMS verification module to verify the signature certificate, certificate chain, and timestamp certificate. After the verification is passed, the public key is used to decrypt the digital signatures to verify the integrity of the software packages.

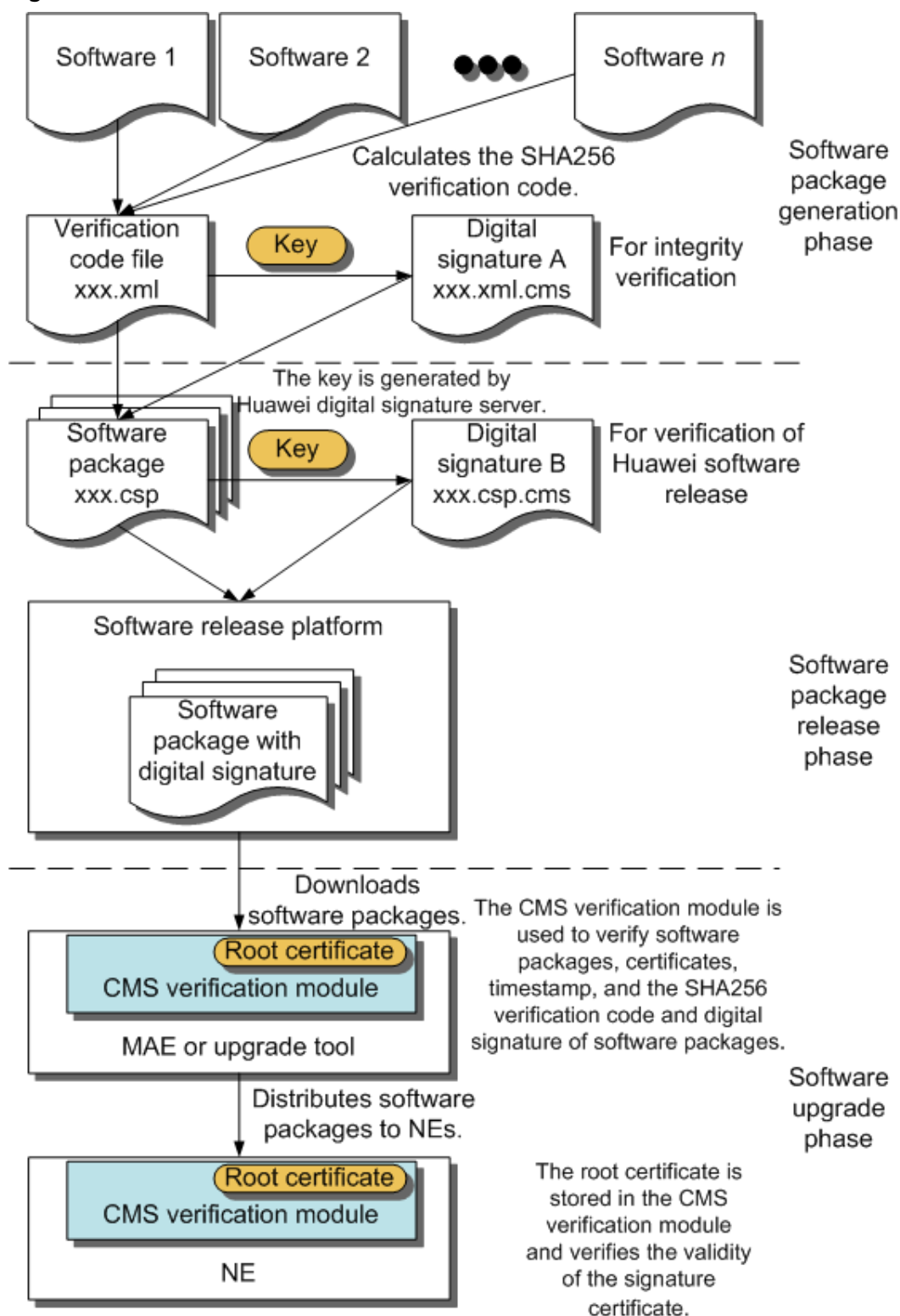
 NOTE

A signature certificate is issued by Huawei Certification Authority (CA) and is used to generate digital signatures for software packages.

Generally, private keys stored on Huawei digital signature server will not be cracked or disclosed. However, to mitigate the risk of private key disclosure, CRL files need to be updated. For details, see [4.1.6.3.4 Possible Issues](#).

[Figure 4-10](#) shows the procedure of the Huawei PKI-CMS solution.

Figure 4-10 Huawei PKI-CMS solution



 NOTE

The OMU is responsible for installing the software package of the base station controller of GSM and UMTS networks. External attackers or unauthorized internal users may tamper with the base station controller software after the OMU software is installed. Therefore, the base station controller checks the integrity of the software on the OMU and reports only one ALM-20723 File Loss or Damage if one or more files are damaged or lost. This alarm is cleared after all the damaged or lost files are restored.

For an OS upgrade, the MAE or upgrade tool checks the integrity of the OS upgrade package.

For an OS driver upgrade, the driver upgrade tool checks the integrity of the OS drive package.

4.1.6.3.4 Possible Issues

Background Information

Each certificate has a validity period. After a certificate is revoked, it becomes invalid. A certificate revocation list (CRL) file lists certificates that are considered as invalid by certificate issuers. Generally, the update period of Huawei CRL files is two months.

Fault Description

If the private key of a PKI-CMS digital signature is leaked out, Huawei will urgently release the latest CRL file to revoke the leaked certificate, preventing NEs from being installed with malicious software. Urgent CRL file release is not required during route maintenance but only required when a private key leaks out. Contact Huawei engineers to perform urgent CRL file release.

For an eGBTS/NodeB/eNodeB/gNodeB

- Step 1** Download the latest CRL file from <https://support.huawei.com/additionalres/pki>, and run the **ULD CERTFILE** command to upload it to the FTP server.
- Step 2** Run the MML command **DLD GENFILE** with **TYPE** set to **SWSCRL** to download the CRL file to the base station.
- Step 3** Run the MML command **DSP SWSCRL** to check whether the CRL file has been updated successfully. [Figure 4-11](#) shows an example of the expected command output.

Figure 4-11 Updating the CRL file for an eGBTS/NodeB/eNodeB/gNodeB

```
To display the software signature certificate revocation list (CRL), run the following command:
DSP SWSACL:;
The result is as follows:
+++ 0 2016-03-07 17:33:56
O&M #456
%%DSP SWSACL:;%%
RETCODE = 0 Operation succeeded.

Software Signature CRL's Details
-----
Issuer Effective Date
C=CN,O=Huawei,CN=Huawei Root CA 2015-10-19 15:38:25
C=CN,O=Huawei,CN=Huawei Code Signing Certificate Authority 2016-3-1 16:54:12
C=CN,O=Huawei,CN=Huawei Timestamp Certificate Authority 2016-3-1 16:56:22
(Number of results = 3)

--- END
```

----End

For a GBTS

- Step 1** Download the latest CRL file from <https://support.huawei.com/additionalres/pki>, and run the **ULD CERTFILE** command to upload it to the FTP server.
- Step 2** Run the MML command **DLD SWSACL** to download the CRL file to the base station controller.
- Step 3** Run the MML command **LOD BTSSWSACL** to load the CRL file to the GBTS.
- Step 4** Run the MML command **DSP BTSSWSACL** to check whether the CRL file has been updated successfully. **Figure 4-12** shows an example of the expected command output.

Figure 4-12 Updating the CRL file for a GBTS

```
+++ admin 2015-04-16 15:02:22
O&M #40643
%%DSP BTSSWSACL: IDTYPE=BYID, BTSID=15;%%
RETCODE = 0 Execution succeeded.

Display BTS CRL's Details
-----
Issuer Effective Date
C=CN,O=Huawei,CN=Huawei Root CA 2015-10-19 7:38:25
C=CN,O=Huawei,CN=Huawei Code Signing Certificate Authority 2016-4-5 8:27:5
C=CN,O=Huawei,CN=Huawei Timestamp Certificate Authority 2016-3-1 8:56:22
(Number of results = 3)

--- END
```

----End

NOTICE

If the CRL file is not replaced in time, use the OMStar-based centralized security management process to check whether the base station on the live network experiences exceptions (for example, the base station is upgraded or the base station traffic is abnormal during the period from the time the private key leaks out to the time the CRL file is updated). If any exception occurs, upgrade the base station to a secure version.

4.1.6.4 Configuration Security Management

Due to the dynamic development of cryptography, many algorithms/protocols that used to be secure can no longer serve the purpose and are therefore not recommended. That is, you are not advised to use the parameter values indicating such weak algorithms, which are described in [4.4.6.4.2 Data Configuration](#). The configuration security management function manages these parameter values for security purposes.

Security management on these parameter values is implemented by using the **SYSTEMOM.CFGSECLEV (LTE eNodeB, 5G gNodeB)** parameter, which specifies the configuration security level. The values and corresponding impacts are as follows:

- **LEVEL0**: There is no restriction on the configurations of algorithm parameters. All parameter values can be used.
- **LEVEL1**: Parameter values indicating weak algorithms cannot be used. The values of the algorithm parameters cannot be changed from those indicating strong algorithms to those indicating weak algorithms.

The configurations in different scenarios are as follows:

- New base station deployment: By default, the value **LEVEL0** is used, and parameter values indicating weak algorithms can be used.
- System upgrade: If **SYSTEMOM.CFGSECLEV (LTE eNodeB, 5G gNodeB)** is configured before the upgrade, the NE inherits the configuration security level after the upgrade. Otherwise, the configuration security level is **LEVEL0**.
- Configuration security level modification: When parameter values indicating weak algorithms are used, the value of **SYSTEMOM.CFGSECLEV (LTE eNodeB, 5G gNodeB)** cannot be changed from **LEVEL0** to **LEVEL1**. If such modification is required, you need to change all the parameter values indicating weak algorithms to those indicating strong algorithms before setting **SYSTEMOM.CFGSECLEV (LTE eNodeB, 5G gNodeB)** to **LEVEL1**.

4.2 Network Analysis

4.2.1 Benefits

This function is used to ensure O&M security.

4.2.2 Impacts

None

4.3 Requirements

4.3.1 Licenses

None

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

None

4.3.2.2 Mutually Exclusive Functions

None

4.3.2.3 Function Impacts

None

4.3.3 Hardware

Base Station Models

RAT	Base Station Model
GSM	3900 and 5900 series base stations (macro base stations)
UMTS	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite
LTE	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite

RAT	Base Station Model
NR	3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910. - DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910. Only NR TDD is supported.

Boards

SSH-encrypted transmission is supported only by the Book BBU and UMPTe/UMPTg/UMPTga/UMPTba/UMPTb series main control boards. Other functions are supported by all main control boards.

RF Modules

This function does not depend on RF modules.

4.3.4 Others

None

4.4 Operation and Maintenance

4.4.1 User Management

4.4.1.1 Data Configuration

4.4.1.1.1 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

User Rights Control

Users are added in either of the following scenarios:

- Adding users of a predefined level, including Administrator(s), Operator(s), User(s), and Guest(s)

Fixed rights have been allocated to such users in their command groups and cannot be changed.

- Adding users of the Custom(s) level

The rights of these users need to be specified in a separate command group.

The following provides configuration examples.

- To add a user of a predefined level, for example, Operator(s), perform the following step:
Run the **ADD OP** command to add an Operator user.
- To add a user of the Custom(s) level who has the rights to use the G_22 command group including the **COL LOG** command so that the user can collect log files, perform the following steps:
 - a. Run the **SET CCGN** command to configure G_22 as the command group.
 - b. Run the **ADD CCG** command to add commands to the G_22 command group. In this step, add the **COL LOG** command to the command group.
 - c. Run the **ADD OP** command to add a user of the Custom(s) level and configure the rights of the G_22 command group.
- To configure the rights of the Custom(s) user to use the file manager, perform the following steps:
 - a. On the WebLMT, click **User-defined command Group** to add commands and function items to a specific command group.
 - b. Run the **ADD OP** or **MOD OP** command to add a Custom(s) user. In this step, set **Command Group** to the command group number specified in the previous step.

NOTE

The changed file manager right settings take effect when the user logs in to the WebLMT the next time.

Password Security Policy

Run the **SET PWDPOLICY** command to set the password policy for local WebLMT users.

FTP User Management

- To configure FTP clients to use encrypted transmission, perform the following step:

Run the **SET FTPSCLT** command with **FTPSCLT.ENCRYMODE** or **FTPCLT.ENCRYMODE** set to **ENCRYPTED**.

NOTE

- An FTP client refers to a module that has the FTP client function on the OMU. The **SET FTPSCLT** command takes effect on all FTP clients.
- If the **FTPSCLT.SSLCERTAUTH** or **FTPCLT.SSLCERTAUTH** parameter is set to **Yes**, a digital certificate must be configured for the connected server. Otherwise, file upload and download will fail. For instructions on how to configure digital certificates when the MAE functions as the FTP server, choose **Security Management > Data Management > Configuring Digital Certificates** in the MAE online help.

- To configure the FTP server to use encrypted transmission, perform the following steps:
 - a. Run the **SET FTPSSRV** command with **FTPSSRV.ENCYRMODE** set to **ENCRYPTED**.
 - b. Reset the ftp_server module for the encrypted transmission mode to take effect.
 - i. Run the **DSP OMU** command to query the OMU mode. If only one result for **Operational state** is displayed, the OMU works in standalone mode. If two results for **Operational state** are displayed, the OMUs work in active/standby mode.
 - ii. Run the **RST OMUMODULE** command to reset the ftp_server module on the active OMU. In this step, set **MNAME** to **ftp_server**. If the OMU works in standalone mode, the encrypted transmission mode takes effect after you perform this step. If the OMU works in active/standby mode, go to the next step.
 - iii. Run the **RST OMUMODULE** command to reset the ftp_server module on the standby OMU. In this step, set **MNAME** to **ftp_server**.
- To configure the port for transmitting data over FTP, perform the following step:

Run the **SET FTPSSRV** command to set the value range of ports for transmitting data over FTP. In this step, set **FTPSSRV.ACDPORTLWLT** and **FTPSSRV.ACDPORTUPLT** to appropriate values.

4.4.1.1.2 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.1.2 Activation Verification

None

4.4.1.3 Network Monitoring

None

4.4.2 OMCH Security

OMCHs are secured using SSL. For details, see *SSL*.

4.4.2.1 OMU Anti-Attack for Base Station Controllers

4.4.2.1.1 When to Use

OMU anti-attack is supported only by base station controllers. The IPTable function of the OS is used to implement OMU anti-attack.

Configuring the whitelist and blacklist for the IPTable function has high risks. To ensure the normal operation of the NE, do not configure the whitelist or blacklist if the network runs properly.

4.4.2.1.2 Data Configuration

Activation

Manage the OMU locally or remotely using PuTTY. Run the DOPRA Linux command **iptables -A INPUT -s restricted IP -i Ethernet adapter -p transport protocol --dport restricted port -j DROP**. [Table 4-11](#) describes parameter settings in this command.

Table 4-11 iptables command parameters

Parameter Name	Description
restricted IP	Set restricted IP to an IP address from which access is denied or allowed. The IP address can be a single IP address or a network segment IP address.
Ethernet adapter	Set Ethernet adapter to the external network adapter of the OMU.
transport protocol	Set transport protocol to TCP or UDP. This parameter is used with restricted port .
restricted port	Set restricted port to the port over which access is prohibited. If you do not specify the -p transport protocol and --dport restricted port parameters, access over all ports is prohibited.

The following is a command example used to allow only users in the 10.141.148.0 network segment to access the WebLMT:

```
iptables -A INPUT -s ! 10.141.148.0/255.255.255.0 -i bond1 -p tcp --dport 80 -j DROP
```

 NOTE

"!" is a logical negation operator.

Deactivation

1. Manage the OMU locally or remotely using PuTTY. Run the DOPRA Linux command **iptables -D INPUT -s restricted IP -i Ethernet adapter -p transport protocol --dport restricted port -j DROP**.
2. Run the DOPRA Linux command **iptables -L** to query all filtering criteria on the OMU. Verify that the new criteria have been removed successfully.

Configuration example:

```
iptables -D INPUT -s ! 10.141.148.0/255.255.255.0 -i bond1 -p tcp --dport 80 -j DROP
```

4.4.2.1.3 Activation Verification

Log in to the PC whose IP address has been restricted. Run the DOPRA Linux command **iptables -L** to query all filtering criteria on the OMU. Verify that the new criteria have been added successfully.

- If port 80 is prohibited, you cannot access the WebLMT. In this situation, check whether you can access the WebLMT on the PC.
- If port 22 is disabled, the OMU cannot be remotely managed. In this case, check whether the OMU can be remotely managed by using the PuTTY on the PC whose IP address is restricted.
- If port 21 is prohibited, you cannot access the ftp_server module on the OMU. In this situation, check whether you can access the ftp_server module on the OMU using an FTP client on the PC.

4.4.2.1.4 Network Monitoring

None

4.4.2.2 SSH-Encrypted Transmission

4.4.2.2.1 When to Use

Scenario 1: automatic activation. If the SSL connection of the OM channel is interrupted and fails to be established for three consecutive times, the SSH port (6011) is automatically enabled to allow login to the SSH server using the SSH client for local O&M.

Scenario 2: manual activation. In scenarios other than scenario 1, if the SSH client needs to be used for local O&M, the administrator can run the **SET SSHCONF** command to enable the SSH port.

NOTE

- The commonly used SSH clients PuTTY (recommended version: 0.75 or later) and OpenSSH (recommended version: 9.5 or later) are supported.
- One main control board supports only one SSH session, and only the admin user can log in to the SSH server.

4.4.2.2.2 Data Configuration

Scenario 1

Activation: automatic activation.

Deactivation: The SSH port is automatically disabled after the SSL connection of the OM channel is restored.

Scenario 2

The following two activation methods can be used:

- Method 1: Run the MML command with the default SSH configuration policy (whose policy ID is 0) to enable the SSH port.

Configuration example:

```
SET SSHCONF: PORTFLAG=ENABLE;
```

- Method 2: Customize an SSH configuration policy. For example, you can add an SSH configuration policy whose policy ID is 1, key exchange algorithm is DH_GROUP14_SHA256, encryption algorithm is AES_CTR_256, HMAC algorithm is HMAC_SHA2_512, and host key algorithm is ECDSA_SHA2_NISTP521.

Configuration example:

```
ADD SSHPOLICY: POLICYID=1,  
KEYEXCHANGEALG=ECDH_SHA2_NISTP256-0&ECDH_SHA2_NISTP384-0&ECDH_SHA2_NISTP521-0&  
DH_GROUP14_SHA256-1, CIPHERALG=AES_CTR_128-0&AES_CTR_256-1,  
HMACALG=HMAC_SHA2_256-0&HMAC_SHA2_512-1,  
HOSTKEYALG=SSH_RSA-0&ECDSA_SHA2_NISTP256-0&ECDSA_SHA2_NISTP384-0&ECDSA_SHA2_NISTP  
521-1;
```

Example of referencing a user-defined SSH policy:

```
SET SSHCONF: POLICYID=1;
```

Deactivation: Run an MML command to disable the SSH port.

Configuration example:

```
SET SSHCONF: PORTFLAG=DISABLE;
```

4.4.2.2.3 Activation Verification

None

4.4.2.2.4 Network Monitoring

None

4.4.3 Security Configuration Check

4.4.3.1 When to Use

The default periodic security configuration check task is performed based on the default template. The default template involves only basic security configuration rules. Operators can add check templates based on their security requirements to identify risks and rectify faults in a timely manner. For example, they can do so in the following scenarios:

- When the default template does not meet their security requirements
- When different NE versions on the live network have different security requirements
- In specific application scenarios, such as IPsec and digital certificates
- When the NE is of an earlier version, whose original configuration may include insecure configuration items

4.4.3.2 Data Configuration

The security configuration check function is used to check and evaluate the security configurations of NEs and to identify risks. It does not involve MML commands or parameter settings. You are advised to rectify the identified risky security configurations by referring to *Security Configuration Hardening Guide* and

Security Configuration Baseline in 3900 & 5900 Series Base Station Product Documentation.

Activating a Check Task

The default check task is enabled by default. To activate the default check task when it is suspended, choose **Access > Security > Security Configuration Management > Security Configuration Check** on the MAE, choose **Task Management > Check Task Management** on the displayed page, and select the task. The task period can be modified.

To create a check task based on your security requirements, choose **Access > Security > Security Configuration Management > Security Configuration Check** on the MAE, choose **Task Management > Check Task Management** on the displayed page, and click **Create**. After the task is created, click **Run** to execute the task. A one-time task is executed once. A periodic task is executed at the specified interval.

Deactivating a Check Task

Choose **Access > Security > Security Configuration Management > Security Configuration Check** on the MAE, and choose **Task Management > Check Task Management** on the displayed page. A one-time task automatically stops after being executed once. A periodic task can be deactivated by suspending it.

4.4.3.3 Activation Verification

None

4.4.3.4 Network Monitoring

None

4.4.4 Application Security

4.4.4.1 Web Security

4.4.4.1.1 When to Use

Web applications are vulnerable to attacks, and therefore HTTPS security policies need to be enabled.

4.4.4.1.2 Data Configuration

Data Preparation

Table 4-12 Parameters for modifying the WebLMT login/access policy on the base station side

Parameter Name	Parameter ID	Setting Notes
TLS/SSL Version	WEBLMT.SSLVER (5G gNodeB, LTE eNodeB)	Set this parameter to its recommended value.
Session Idle Timeout	WEBLMT.IDLETIMEOUT (5G gNodeB, LTE eNodeB)	The value 10 is recommended.
Session Maximum Lifetime	WEBLMT.MAXLIFETIME (5G gNodeB, LTE eNodeB)	The value 480 is recommended.

Table 4-13 Parameters for setting LMT login and transmission policies on the BSC6900/BSC6910 side on GSM and UMTS networks

Parameter Name	Parameter ID	Setting Notes
Policy for login to LMT and transmission	WEBLOGINPOLICY.POLICY	Set this parameter to its recommended value.
TLS/SSL Version	WEBLOGINPOLICY.SSLVER	Set this parameter to its recommended value.

Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

- To configure the WebLMT login policy for the BSC6900/BSC6910 on GSM and UMTS networks, perform the following steps:

```
SET WEBLOGINPOLICY: SSLVER=TLSV12-1&TLSV13-1;  
//Restarting the WebLMT server for the configured login policy to take effect  
RST OMUMODULE: TG=ACTIVE, MNAME=weblmt;
```

NOTE

When the WebLMT server restarts, WebLMT clients are disconnected and therefore cannot receive the restart command response from the WebLMT server. In addition, an error message indicating that the command fails to be sent is displayed. Ignore this error prompt because the command was successfully sent.

- To configure the WebLMT login policy for the base station, perform the following step:

```
SET WEBLOGINPOLICY: SSLVER=TLSV12-1&TLSV13-1, IDLETIMEOUT=10, MAXLIFETIME=480;
```

Using the MAE-Deployment

Security policy level configuration on the MAE-Deployment can be used to configure the WebLMT login policy for existing base stations.

You can perform consistency check on the **Current Area** on the MAE-Deployment. If security policies also need to be delivered for synchronization according to the check results, create or select a planned area first.

- Step 1** On the MAE-Deployment, choose **Advanced > Data Management > Consistency Check > Security Policy Level** to set the consistency check parameters for security policies.
- Step 2** Select the NEs for which consistency check is to be performed, execute the check to generate a check report.
- Step 3** Based on the check report, correct the configurations on NEs in batches in the event of inconsistency.

----End

4.4.4.1.3 Activation Verification

None

4.4.4.1.4 Network Monitoring

None

4.4.4.2 Time Security

Correct time synchronization guarantees normal operation of O&M systems. A standalone NTP server needs to be configured and wireless NEs function as NTP clients. NTP security policies ensure correct time synchronization. The NTP server is generally configured by operators and therefore the NTP security policies on wireless NEs are configured based on the interworking requirements of the NTP server.

4.4.4.2.1 SNTP Security for the BSC6900/BSC6910 (GSM/UMTS)

Data Configuration

Run the **ADD SNTPSRVINFO** command to add the IP address and port number for the SNTP server on the base station controller and set the SNTP time

synchronization security policy. You can run the **LST SNTPCLTTPARA** command to query information about the SNTP server.

Activation Verification

NTP security is activated if the NTP parameters are correctly configured and NTP link status is normal.

Network Monitoring

None

4.4.4.2.2 Deployment of NTP Security Authentication for the Base Station/BSC6960

Data Preparation

Table 4-14 describes key parameters that must be set in the **NTPCP** MO to activate NTP security authentication.

Table 4-14 Data to be prepared before activating NTP security authentication

Parameter Name	Parameter ID	Setting Notes
IPv4 Address of NTP Server	NTPCP.IP (<i>5G gNodeB, LTE eNodeB</i>)	This parameter specifies the IPv4 address of the NTP server.
Port Number	NTPCP.PORT (<i>5G gNodeB, LTE eNodeB</i>)	This parameter specifies the number of the time synchronization port on the NTP server. The NTP client synchronizes with the NTP server through the specified port.
Synchronization Period	NTPCP.SYNCCYC LE (<i>5G gNodeB, LTE eNodeB</i>)	This parameter specifies the NTP time synchronization interval.
Authentication Mode	NTPCP.AUTHMO DE (<i>5G gNodeB, LTE eNodeB</i>)	This parameter specifies the NTP authentication mode.
Authentication Key	NTPCP.KEY (<i>5G gNodeB, LTE eNodeB</i>)	This parameter specifies the key used for NTP authentication.
Authentication Key Index	NTPCP.KEYID (<i>5G gNodeB, LTE eNodeB</i>)	This parameter specifies the index of the authentication key on the NTP server. The local index must be the same as that on the NTP server.

Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After

Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

```
//Configuring an NTP client  
ADD NTPC: MODE=IPV4, IP="192.168.88.168", PORT=123, SYNCCYCLE=10, AUTHMODE=PLAIN;
```

Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

Activation Verification

The activation observation procedure is as follows:

- Step 1** Run the **LST NTPC** command to query the NTP configuration information. Verify that the parameter settings in the command output are consistent with that configured in the activation procedure.
- Step 2** Run the **DSP NTPC** command to query the time synchronization information of the base station. Verify that the value of **Link State of Current NTP Server** is **Available** in the command output.
- Step 3** Run the **LST LATESTSUCCDATE** command to query the latest successful time synchronization of the base station. Verify that the value of **Latest Successful Synchronization Time** is the same as the time that time synchronization was recently performed.

----End

If all the preceding verifications are true, NTP security authentication is activated.

Network Monitoring

None

4.4.5 Security Monitoring

4.4.5.1 Security Monitoring

Security monitoring is used on the MAE to monitor security status of devices on the live network. No engineering guidelines are involved.

4.4.5.2 Security Alarms, Events, and Logs

Security alarms, events, and logs are always enabled and do not involve engineering guidelines.

4.4.6 Data Security

4.4.6.1 User Data Pseudonymization

4.4.6.1.1 When to Use

- GSM and UMTS
Wireless networks use hash algorithms to generate pseudonyms to replace user identities involved in OM to protect user privacy. For details, see *User Data Pseudonymization* in GBSS feature documentation or RAN feature documentation.
- LTE
 - If the SC algorithm switch is turned on, a pseudonymity key must be configured on the MAE.
 - The setting of the **STMSI_MASK_SW** option depends on the operator's management policy. If this option is selected, S-TMSI mask is enabled. With this function, S-TMSIs in internal and external CHRs are replaced by all 0s. If this option is deselected, this function is disabled.
- NR
The setting of the **STMSI_MASK_SW** option depends on the operator's management policy. If this option is selected, S-TMSIs are not recorded in CHRs. If this option is deselected, S-TMSIs can be recorded in CHRs as required.

4.4.6.1.2 Data Configuration

Data Preparation

[Table 4-15](#) lists the data to be prepared for function activation on the eNodeB.

Table 4-15 Parameters for controlling S-TMSI anonymization in CHR on the eNodeB

Parameter Name	Parameter ID	Description
CHR Output Control Switch	eNodeBChrOutputCtrl.ChrOutputCtrlSwitch	The setting of the STMSI_MASK_SW option depends on the operator's management policy. If this option is selected, S-TMSI mask is enabled. With this function, S-TMSIs in internal and external CHR are replaced by all 0s. If this option is deselected, this function is disabled.

Table 4-16 lists the data to be prepared for function activation on the gNodeB.

Table 4-16 Parameters for disabling S-TMSI recording in CHR on the gNodeB

Parameter Name	Parameter ID	Description
CHR Switch	gNodeBParam.ChrSwitch	The setting of the STMSI_MASK_SW option depends on the operator's management policy. If this option is selected, S-TMSIs are not recorded in CHR. If this option is deselected, S-TMSIs can be recorded in CHR as required.

Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

- Enabling user data pseudonymization on the eNodeB
//Turning on the S-TMSI mask switch for CHR output control on the eNodeB
MOD ENODEBCHROUTPUTCTRL: ChrOutputCtrlSwitch=STMSI_MASK_SW-1;
- Enabling user data pseudonymization on the gNodeB
//Turning on the switch for disabling S-TMSI recording in CHR on the gNodeB
MOD GNODEBPARAM: ChrSwitch=STMSI_MASK_SW-1;

Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.6.1.3 Activation Verification

None

4.4.6.1.4 Network Monitoring

None

4.4.6.2 Security Management of Configuration Files

4.4.6.2.1 When to Use

You are advised to encrypt a configuration file in the following two scenarios:

- Offline transmission of a configuration file
- Online permanent storage of a configuration file

4.4.6.2.2 Data Configuration

Data Preparation

The following table lists MML commands used for configuration file encryption.

Table 4-17 MML commands used for configuration file encryption

MML Command	Operation Type	Parameter Description
DLD BATCHFILE	Import	ENCRYPTMODE : Encryption mode of a configuration file. The value can be UNENCRYPTED or PWD_ENCRYPTED . FILEPWD : Password used for encrypting a configuration file. The value consists of 6 to 64 digits.
DLD CFGFILE	Import	
RUN BATCHFILE	Import	
RTR DB	Import	
BKP CFGFILE	Export	

MML Command	Operation Type	Parameter Description
ULD CBCFGFILE	Export	
BKP DB	Export	
EXP CFGMML	Export	
EXP CFGBCP	Export	

Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

On the WebLMT, run MML commands listed in [Data Preparation](#) to encrypt a configuration file.

Using the MAE-Deployment

Perform the following steps related to configuration file encryption on the MAE or WebLMT:

- On the MAE, select the encryption option in the window for manual or automatic data backup.
- Select the encryption option when the MAE is generating a configuration file.
- On the WebLMT, browse and activate the encrypted configuration file.

4.4.6.2.3 Activation Verification

- When a configuration file is exported, check whether the configuration file is encrypted by observing the file name extension. If a configuration file is encrypted, the file name is suffixed with .ecf. For example, the file name changes from NodeB.xml to NodeB.xml.ecf after encryption.
- When an encrypted configuration file is imported, you can execute or browse the original configuration file after entering the correct password.

4.4.6.2.4 Network Monitoring

None

4.4.6.3 Digital Signature-based Software Integrity Protection

This function is enabled by default. Function deployment is not required.

4.4.6.4 Configuration Security Management

4.4.6.4.1 When to Use

O&M personnel need to ensure that all parameter values indicating weak algorithms are not and will not be used before setting **SYSTEMOM.CFGSECLEV (LTE eNodeB, 5G gNodeB)** to **LEVEL1**. If a parameter value indicating a weak algorithm is still in use, the value of **SYSTEMOM.CFGSECLEV (LTE eNodeB, 5G gNodeB)** cannot be changed from **LEVEL0** to **LEVEL1**.

4.4.6.4.2 Data Configuration

Data Preparation

Table 4-18 lists the parameters that must be set for configuration security management.

Table 4-18 Parameters that must be set for configuration security management

Parameter Name	Parameter ID	Description
Configuration Security Level	SYSTEMOM.CFGSECLEV (LTE eNodeB, 5G gNodeB)	Indicates the configuration security level. This parameter is used to manage the parameter values indicating weak algorithms. If this parameter is set to LEVEL0 , all parameter values can be used. If this parameter is set to LEVEL1 , parameter values indicating weak algorithms cannot be used. The default value is LEVEL0 .

Table 4-19 lists the parameter values that cannot be used when the **SYSTEMOM.CFGSECLEV (LTE eNodeB, 5G gNodeB)** parameter is set to **LEVEL1**.

Table 4-19 Parameter values indicating weak algorithms

Category	Parameter Name	Parameter ID	Parameter Value That Cannot Be Used After Being Disabled
IPsec-related algorithms	Encryption Algorithm	IKEPROPOSAL.ENCALG (LTE eNodeB, 5G gNodeB)	DES, 3DES
	Authentication Algorithm	IKEPROPOSAL.AUTHALG (LTE eNodeB, 5G gNodeB)	MD5, SHA1, AES_XCBC_96

Category	Parameter Name	Parameter ID	Parameter Value That Cannot Be Used After Being Disabled
	Diffie-Hellman Group	IKEPROPOSAL.DHGRP (LTE eNodeB, 5G gNodeB)	DH_GROUP1, DH_GROUP2
	PRF Algorithm	IKEPROPOSAL.PR FALG (LTE eNodeB, 5G gNodeB)	HMAC_MD5, HMAC_SHA1, AES128_XCBC
	RSA Padding	IKEPROPOSAL.RS APADDING (5G gNodeB, LTE eNodeB)	PKCS1V15
	Version	IKEPEER.IKEVERSION (LTE eNodeB, 5G gNodeB)	IKE_V1
	AH Authentication Algorithm	IPSECPROPOSAL.AHAUTHALG (LTE eNodeB, 5G gNodeB)	MD5, SHA1, AES-XCBC-MAC-96
	ESP Authentication Algorithm	IPSECPROPOSAL.ESPAUTHALG (LTE eNodeB, 5G gNodeB)	MD5, SHA1, AES-XCBC-MAC-96
	ESP Encryption Algorithm	IPSECPROPOSAL.ESPENCALG (LTE eNodeB, 5G gNodeB)	DES, 3DES
	Perfect Forward Secrecy	IPSECPOLICY.PFS (LTE eNodeB, 5G gNodeB)	PFS_GROUP1, PFS_GROUP2
	IKE Version	SECURITYTEMPLATE.IKEVERSION (LTE eNodeB, 5G gNodeB)	IKE_V1
	Perfect Forward Secrecy	SECURITYTEMPLATE.PFS (LTE eNodeB, 5G gNodeB)	PFS_GROUP1, PFS_GROUP2
PKI-related algorithms	Signature Algorithm	CERTREQ.SIGNALG (LTE eNodeB, 5G gNodeB)	SHA1 (not applicable to the BSC6960)

Category	Parameter Name	Parameter ID	Parameter Value That Cannot Be Used After Being Disabled
	Key Size	CERTREQ.KEYSIZE <i>(5G gNodeB, LTE eNodeB)</i>	KEYSIZE1024 (not applicable to the BSC6960)
	Signature Algorithm	CA.SIGNALG <i>(LTE eNodeB, 5G gNodeB)</i>	SHA1 (not applicable to the BSC6960)
	Certificate Request Signature Algorithm	CA.CERTREQSIGN ALG <i>(LTE eNodeB, 5G gNodeB)</i>	SHA1 (not applicable to the BSC6960)
	Key Size	CA.KEYSIZE <i>(LTE eNodeB, 5G gNodeB)</i>	KEYSIZE1024 (not applicable to the BSC6960)
	TLS Secure Cipher Suite Switch	CA.TLSSECCSSW <i>(LTE eNodeB, 5G gNodeB)</i>	OFF
	Minimum DH Length	CRLTSK.MINDHLEN <i>(LTE eNodeB, 5G gNodeB)</i>	DH_0, DH_512, DH_1024 (not applicable to the BSC6960)
	Authenticate Peer	CRLTSK.AUTHPEER <i>(5G gNodeB, LTE eNodeB)</i>	DISABLE
	TLS Secure Cipher Suite Switch	CRLTSK.TLSSECCSSW <i>(LTE eNodeB, 5G gNodeB)</i>	OFF
	TLS Secure Cipher Suite Switch	FTPCLT.TLSSECCSSW <i>(LTE eNodeB, 5G gNodeB)</i>	OFF
	Support SSL Certificate Authentication	FTPCLT.SSLCERT AUTH FTPCLT.SSLCERT AUTH <i>(LTE eNodeB, 5G gNodeB)</i>	No
	Minimum DH length of FTPS	FTPCLT.MINDHLEN <i>(LTE eNodeB, 5G gNodeB)</i>	DH_0, DH_512, DH_1024 (not applicable to the BSC6960)
FTP-related algorithms	TLS Secure Cipher Suite Switch	FTPCLT.TLSSECCSSW <i>(LTE eNodeB, 5G gNodeB)</i>	OFF

Category	Parameter Name	Parameter ID	Parameter Value That Cannot Be Used After Being Disabled
	Support SSL Certificate Authentication	FTPSCLT.SSLCERT AUTH FTPCLT.SSLCERTA UTH (LTE eNodeB, 5G gNodeB)	No
	Minimum DH length of FTPS	FTPCLT.MINDHLEN (LTE eNodeB, 5G gNodeB)	DH_0, DH_512, DH_1024 (not applicable to the BSC6960)
NTP-related algorithms	Authentication Mode	NTPCP.AUTHMODE (5G gNodeB, LTE eNodeB)	DES_S, DES_N, DES_A, MD5

Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

```
SET SECPY: CFGSECLEV=LEVEL1;
```

Deactivation Command Examples

```
SET SECPY: CFGSECLEV=LEVEL0;
```

Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.6.4.3 Activation Verification

This function is enabled by default. You can run the **LST SECPY** command to query the current security level.

4.4.6.4.4 Network Monitoring

None

5 Parameters

The following hyperlinked EXCEL files of parameter documents match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- eNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the product documentation delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter, which may be only a bit of a parameter. View its information, including the meaning, values, impacts, and product version in which it is activated for use.

----End

6 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- eNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference for the software version used on the live network from the product documentation delivered with that version.

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

Step 1 Open the EXCEL file of performance counter reference.

Step 2 On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.

Step 3 Click **OK**. All counters related to the feature are displayed.

----End

7 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

8 Reference Documents

- *SSL*
- *PKI*
- *SRAN Networking and Evolution Overview*
- *User Data Pseudonymization* in GBSS feature documentation or RAN feature documentation
- *GBTS Equipment and OM Security* in GBSS feature documentation
- *3900 & 5900 Series Base Station MML Command Reference* in *3900 & 5900 Series Base Station Product Documentation*
- *Log Management* in MAE product documentation
- Sections related to security configuration check in MAE product documentation
- *Security Configuration Hardening Guide* in *3900 & 5900 Series Base Station Product Documentation*
- *Security Configuration Baseline* in *3900 & 5900 Series Base Station Product Documentation*